Notebook I — Conventions and Setup: Quantum Screening Doublets for the Symmetric Carrollian W_3 Algebra

Contents

Conventions, Free Fields, and the Precise Setup of the Carrollian Screening Problem **1**

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Research Notebook · Volume I of IV

Conventions, Free Fields, and the Precise Setup of the Carrollian Screening Problem

Preparatory notebook for Quantum Screening Doublets for the Symmetric Carrollian W_3 Algebra

Status setup / conventions-locking **Primary sources** [FH25], [DDF26], [FPT26] **Computation** Python (Pyodide + SymPy) inline **Date** 1 August 2026

0 Purpose, scope, and the audit protocol

This notebook fixes the conventions, signs and normalisations needed for the formal oscillator and parameter calculations of the screening programme, derives the objects the paper's theorems will be about, and itemises the operator-level data deliberately deferred to Notebook II. It is a formal algebraic setup notebook, not a complete operator construction.

The notebook is executable. Python cells run in your browser via Pyodide with SymPy. **What the cells cover**, exactly: the classical Miura polynomial identities; the single-field and general-root screening weights; the parent residue identity of Proposition 3.4; the contracted oscillator brackets on the δ-support; the $N = 2$ contraction of $T, \bar{T}$; the F_0/F_1 parameter hierarchy; the zero-mode half-shift and cylinder mode factor; the formal associated-graded jet; and the projector identities of the index space. **What they do not cover:** source transcription, averaged ordering, contour orientation, cocycles, the even-jet commutant question, and the bulk normalisation dictionary — all of which are prose, inference, or open, and are marked accordingly.

Audit protocol (P0)

The rule, stated so that the document actually satisfies it — this is a load-bearing displayed claim protocol, not an every-box protocol: every load-bearing displayed claim — anything the paper will cite or build on — is marked and appears in the Section 10 ledger. Purely definitional or expository boxes (conventions, definitions, intuition) may be unmarked; they introduce notation rather

than assert results. Marks attach to boxes, not to individual lines; where a box mixes provenance, the surrounding text says which equation is which. The marks are:

- verbatim — faithful transcription from a primary source, with the source's own equation number given: no mathematical change (typographic reformatting and our notation macros are permitted).
- derived — obtained here from verbatim input by a computation shown in full. Reproducible by the accompanying Python cell.
- imported — standard material from the wider literature, translated into our locked conventions. The translation is always shown explicitly, never assumed.
- rewrite — a convention-preserving algebraic rewrite of source material (e.g. cylinder modes to plane OPE), or a field redefinition. The redefinition is always exhibited.
- formulated here — formulated in this notebook; no prior statement was located in the dated search protocol of §1.1. This is a literature-search status, not a categorical claim of global novelty. Always accompanied by a full justification and an explicit statement of what would falsify it.
- open — a conjecture or an unresolved item. Never used as an input to anything downstream.

A load-bearing claim absent from the ledger is not licensed for the paper. Section 10 collects the ledger, including the zero-mode, ordering, contour and module assumptions that are not themselves displayed equations.

Layer discipline. Six layers are kept apart throughout and are never allowed to borrow each other's authority: (a) classical Poisson/mode identities; (b) quantum normal-ordered OPEs; (c) cylinder versus plane conventions; (d) formal ϵ-jets of commuting symbols; (e) vertex-operator intertwiners on a specified module; (f) Hilbert-space/AQFT representations. Most of what follows lives in (a)–(d). Layer (e) is set up but not completed here; layer (f) is out of scope for Paper 1.

Scope discipline

This notebook does not prove the commutant theorem, and does not close the operator layer. It sets up the problem so that the commutant theorem becomes a finite, well-posed computation at the level of formal densities and parameters. It does **not** close the operator layer: contour orientation, the barred equal-time map, vertex normal ordering, the zero-mode cocycle, multiplicative renormalisations and the logarithmic residue for $\varphi^- e^{\beta\varphi^+}$ are all explicitly deferred (the "Missing data" box of §7.1, D4, §9). **Notebook I is a formal algebraic setup notebook**, and its results should be cited as such. Two things in particular are deliberately deferred: (i) the operator layer and the quantum ordering analysis of the tangent screening, to Notebook II; and (ii) the A_2 computation with $U_3^{\pm}$, to Notebook III. What is settled here — and this was not obvious at the outset — is the leading screening condition F_0, obtained by two independent routes in §7.2.

Running the Python cells The first cell you run will download the Pyodide runtime and SymPy (roughly 10 MB, cached thereafter). Allow a few seconds. A status pill appears in the lower right. All later cells share one interpreter session, so definitions persist between cells; run them in order the first time.

pythonCell 0 — environment and shared symbols

```
import sympy as sp

# --- shared symbolic alphabet used throughout this notebook -------------------
eps      = sp.symbols('epsilon', positive=True)      # contraction parameter
alpha    = sp.symbols('alpha')                       # parent background charge (unbarred copy)
alphab   = sp.symbols('alphabar')                    # parent background charge (barred copy)
ap, am   = sp.symbols('alpha^+ alpha^-')             # Carrollian background charges
b        = sp.symbols('b')                           # TODA COUPLING ONLY (beta_+ = 2b)
beta, betabar = sp.symbols('beta betabar')           # full single-field screening momenta
b1       = sp.symbols('beta_1')                      # leading Carrollian screening momentum
b3, b3b  = sp.symbols('beta_3 beta_3bar')            # subleading screening momenta
z, w     = sp.symbols('z w')
N        = sp.symbols('N', positive=True, integer=True)

print("SymPy", sp.__version__, "ready.")
print("Shared symbols:", [str(s) for s in (eps,alpha,alphab,ap,am,b,beta,betabar,b1,b3,b3b)])
```

1 The programme and the precise gap

1.1 Where this sits: four notebooks, three papers

Programme map (authoritative)

Two sequences must not be confused. The **notebooks** are the working record:

$$\underbrace{\text{I}}_{\text{formal setup}} \rightarrow \underbrace{\text{II}}_{A_1\text{ operator layer}} \rightarrow \underbrace{\text{III}}_{A_2/W_3\text{ commutation}} \rightarrow \underbrace{\text{IV}}_{\text{bulk match, } N=2\text{ anchors}} \rightarrow \text{manuscript}$$

The **papers** are the intended outputs: Paper 1, the screening theorem (this programme); Paper 2, the boundary Carrollian W_3 algebra; Paper 3, the nonlocal screening algebra and its quantum group. Notebook IV may be merged into III only under the conditions recorded in `ROADMAP.md`.

Three recent results define the frontier. Fredenhagen and Hörl[FH25] construct the proper symmetric quantum Carrollian W_N-algebra by contracting two copies of a relativistic W_N-algebra at the level of the Miura operator, and prove that with an averaged ordering prescription the basic structure constants coincide with the classical ones. D'Arcy, Delfante and Fredenhagen[DDF26] derive, from asymptotically flat higher-spin Chern–Simons gravity in a flat diagonal gauge, a classical Carrollian Coulomb gas: celestial free bosons, Carrollian screening charges generating the residual pure-gauge transformations, and Carrollian Miura currents as the invariants. Fredenhagen, Prohazka and Tiefenbacher[FPT26] show that massless Carrollian free fields generically require nonregular states with a nonseparable zero-mode sector.

The gap is precise and is stated in the sources themselves. [FH25] lists as open both the role of screening charges in the Carrollian setting and the construction of representations from the free fields. [DDF26] constructs the screening charges only classically and presents quantisation as the next step. We have found no construction of the quantum screening operators of the symmetric Carrollian algebra as of 1 August 2026, on the basis of the following dated protocol: full-text bibliography searches for the identifiers `2509.14441` and `2604.27068`; INSPIRE/publisher citation trackers on those two papers and on Fateev–Lukyanov; and keyword sweeps on Carrollian screening, Carrollian Coulomb gas, Carrollian Toda, quantum Carrollian W. The sweep is to be repeated at each drafting milestone and on submission day; a null result from a search engine is not a proof of absence, and unpublished work inside the originating group is not visible to it.

The object this programme builds

The quantum operators $Q_i^{(0)}, Q_i^{(1)}$ such that the currents of the proper symmetric quantum Carrollian W_3-algebra of [FH25] lie in their joint commutant,

$$\mathcal{W}_3^{\text{Car}} \subseteq \bigcap_{i,a} \ker\big(\operatorname{ad} Q_i^{(a)}\big),$$

and whose semiclassical limit is the set of four elementary bulk screening charges of [DDF26], eqs. (5.26a–d). **Inclusion, not equality.** Proving that the commutant is exactly $\mathcal{W}_3^{\text{Car}}$ requires a screening-kernel theorem — a graded character comparison, a cohomological (Felder-type) resolution, or a strong-generation argument — none of which is attempted in Paper 1. The semiclassical limit is the $G \to 0$ limit of the bulk normalisation D5, not an $\hbar \to 0$ limit: no $\hbar$ appears anywhere in our oscillator algebra, and the earlier use of that phrase was meaningless.

1.2 The five target statements

Notebooks II–IV and the manuscript will establish the following, in the allocation fixed by `ROADMAP.md`. They are recorded here so that every convention fixed below can be checked against what it is ultimately for.

Label	Statement	Notebook
Theorem 1	Contraction theorem: the elementary A_2 screening doublets $\mathscr{S}_i^{(0)}, \mathscr{S}_i^{(1)}$ are derived as limits of the two parent chiral screening systems, including zero-mode factors, ordering prescription and momentum scaling.	I (jet) → II (A_1 operators) → III (A_2)

Label	Statement	Notebook
Lemma 2	Filtration/truncation lemma: the contraction order of U_k^σ against $\mathscr{S}_i^{(a)}$ is bounded by $\sigma + a \in \{0, 1, 1, 2\}$. Proved abstractly, before any OPE.	I §5.3
Theorem 3	Commutation theorem: $[Q_i^{(a)}, U_{k,n}^\sigma] = 0$ for $k = 2, 3$, $a = 0, 1$, $\sigma = \pm$, as identities of intertwiners between specified free-field modules. Establishes inclusion $\mathcal{W}_3^{\mathrm{Car}} \subseteq \bigcap \ker \operatorname{ad} Q$; equality is not claimed.	III
Theorem 4	Parameter relations: the hierarchy $F_0(\alpha^+, \beta_1) = 0$, $F_1(\alpha^+, \alpha^-, \beta_1, \beta_3, \bar{\beta}_3) = 0$, settling where the relativistic reciprocal quantum relation survives; and the existence of the limiting operators. F_0 is already exact and ordering-independent (Prop. 7.3b); the open part is existence, F_1, and $N = 3$.	I (A_1 parameters) → II (operators) → III (A_2)
Theorem 5	Semiclassical ($G \to 0$) holographic matching to [DDF26] (5.26a–d) at the locus $\alpha^- = 0$, completing the dictionary D5 (root basis, $8\pi G$ vs $6\pi G$, the constants κ_n).	IV
Stretch	Closure of $[Q_i^{(a)}, Q_j^{(b)}]$ onto quantum analogues of the composite nonlocal charges [DDF26] (5.27a–b). Not a submission gate.	Paper 3 horizon

2 The parent relativistic W_N-algebra: conventions locked

The current and Miura conventions of this section are taken verbatim from [FH25], and we adopt their normalisations without modification even where another choice would be more familiar. §2.4 is the exception and is marked as such. **[FH25] do introduce a potential** — their §3.1 states $J_j = i\,\partial\phi_j^{\mathrm{FH}}$ on the cylinder — so what §2.4 supplies is not a potential from nothing but an explicitly named notation bridge,

$$\varphi_j := i\,\phi_j^{\mathrm{FH}}, \qquad \text{so that} \qquad J_j = \partial\varphi_j.$$

The two symbols are kept distinct throughout: ϕ^{FH} is [FH25]'s field, φ is ours, and every reality or Hermiticity statement is made in terms of ϕ^{FH}.

2.1 Free fields and their brackets

Convention C1 — parent currents verbatim [FH25] (3.14), (3.4)

N spin-one currents J_i, $i = 1, \dots, N$, with classical brackets and traceless constraint

$$[J_{i,m}, J_{j,n}]^{\mathrm{cl}} \;=\; m\,\delta_{m,-n}\Big(\delta_{ij} - \tfrac{1}{N}\Big), \qquad J_1 + J_2 + \cdots + J_N = 0.$$

Source note S1 — which equation this is

[FH25] states the free-field bracket twice: at (3.3), in setting up the classical Miura transformation, and again at (3.14), when preparing the contraction. **We cite (3.14)**, and every calculation below uses it.

The reason for being explicit: **equation (3.3) as printed omits the mode factor m, while (3.14) contains it.** This has been confirmed by direct inspection of the canonical arXiv v3 PDF (whose SHA-256 is recorded in the manifest). We record it as a **source erratum note**, not as a new result, and we do not speculate about its cause. It is immaterial either way: the m-carrying form is fixed independently and redundantly by (3.14), by the contracted brackets (3.15), and by the $N = 2$ normalisation (2.10), with which Check 2.1 shows it agrees. Anyone reproducing this notebook should read (3.14).

We abbreviate the pairing matrix as $\kappa_{ij} := \delta_{ij} - \frac{1}{N}$, the projector onto the traceless subspace. **Two distinct statements, kept apart.** The displayed relation is [FH25]'s classical cylinder mode bracket (their $[\,\cdot,\cdot\,]^{\rm cl} = i\{\cdot,\cdot\}$). Separately, we adopt the quantum radial plane OPE

$$J_i(z)J_j(w) \sim \frac{\kappa_{ij}}{(z-w)^2} \qquad \text{(adopted quantum convention, not derived from the line above)}$$

imported — it is [FH25]'s own quantum free-field normalisation (their §2, footnote 3, and §4.1), and it is the convention in which §§3 and 7.2 are computed. **A classical cylinder Poisson bracket does not become a quantum radial OPE by a coordinate change**, and an earlier draft wrongly labelled the passage a "rewrite". The two are used in different sections — OPE form in §§3, 7.2, mode form in §§4–6 — and their exact bridge is deferred (see the scope decision in §6 and Erratum E1).

Convention C2 — the $N = 2$ single-field normalisation verbatim [FH25] (2.10), fn. 3

$$[J_m, J_n] = \tfrac{m}{2}\,\delta_{m,-n}, \qquad \text{equivalently} \qquad J(z)J(w) \sim \frac{1/2}{(z-w)^2}.$$

Consistency check 2.1 derived

C1 and C2 must agree. For $N = 2$ the constraint gives $J_1 = -J_2$; writing $J := J_2$ we get $\kappa_{22} = 1 - \frac{1}{2} = \frac{1}{2}$, hence $[J_m, J_n] = m\kappa_{22}\delta_{m,-n} = \frac{m}{2}\delta_{m,-n}$. Agreement is exact. This is the first of several checks that C1 and the $N = 2$ section of [FH25] are one convention, not two — a fact we will lean on heavily.

2.2 The Miura transformation

Convention C3 — Miura operator verbatim [FH25] (3.1), (3.5), (3.6)

$$\prod_{i=1}^{N} (\alpha\partial - J_i) \;=\; (\alpha\partial)^N - \sum_{k=2}^{N} u_k\,(\alpha\partial)^{N-k},$$

$$ T=u_2=-\sum_{i1}(j-1)\,\partial J_j, \qquad c^{\rm cl}=-(N-1)N(N+1)\alpha^{2}. $$

The product is ordered with increasing i from left to right, and ∂ acts on everything to its right.

Physical intuition — what the Miura map is doing

The Miura transformation is a statement about factorising a differential operator. The N-th order operator $\mathcal{L} = (\alpha\partial)^N - \sum u_k(\alpha\partial)^{N-k}$ is the object whose kernel transforms covariantly; the currents u_k are its coefficients, i.e. its invariants. Writing $\mathcal{L}$ as a product of first-order factors $(\alpha\partial - J_i)$ is a Miura/Darboux factorisation: the J_i are "square roots" carrying the gauge redundancy, and the u_k are what survives. In the Chern–Simons picture of [DDF26] this is literally a gauge choice — the diagonal gauge — with the J_i the diagonal entries and the u_k the gauge invariants. The screening charges will be the generators of the residual redundancy: they move the J_i and fix the u_k. That is the entire logic of the Coulomb gas, and it is why "screening" and "pure gauge" are the same statement viewed from the two sides of the duality.[DDF26]

pythonCell 1 — CLASSICAL differential-operator Miura expansion vs eq. (3.5), N = 2,3,4

```
import sympy as sp
zz = sp.Symbol('z')

def miura_coeffs(n):
    """Expand prod_i (alpha*d - J_i) acting on a test function; return u_k, k=2..n,
```

```
        using the constraint sum_i J_i = 0."""
    a  = sp.Symbol('alpha')
    Js = [sp.Function('J%d'%(i+1))(zz) for i in range(n)]
    f  = sp.Function('f')(zz)
    # apply factors right-to-left:  (a d - J_1)(a d - J_2)...(a d - J_n) f
    expr = f
    for i in reversed(range(n)):
        expr = a*sp.diff(expr, zz) - Js[i]*expr
    expr = sp.expand(expr)
    # subtract (a d)^n f and read off -u_k (a d)^{n-k} f
    us = {}
    rem = sp.expand(expr - a**n*sp.diff(f, zz, n))
    for k in range(1, n+1):
        dk   = sp.diff(f, zz, n-k)
        coef = sp.expand(rem.coeff(dk))
        # strip terms that involve higher derivatives of f
        coef = coef.subs({sp.diff(f, zz, j): 0 for j in range(1, n+1)})
        us[k] = sp.simplify(-coef/a**(n-k))
        rem   = sp.expand(rem - (-us[k])*a**(n-k)*dk)
    return us, Js, a

for n in (2,3,4):
    us, Js, a = miura_coeffs(n)
    # impose the traceless constraint J_1 = -(J_2+...+J_n)
    sub = {Js[0]: -sum(Js[1:])}
    u1  = sp.simplify(us[1].subs(sub))
    u2  = sp.expand(sp.simplify(us[2].subs(sub)))
    # FH (3.5):  T = u_2 = -sum_{i<j} J_i J_j + alpha * sum_{j>1} (j-1) dJ_j
    fh  = -sum(Js[i]*Js[j] for i in range(n) for j in range(i+1,n)) \
          + a*sum((j)*sp.diff(Js[j], zz) for j in range(1,n))
    fh  = sp.expand(sp.simplify(fh.subs(sub)))
    print("N =", n, "|  u_1 (must vanish):", sp.simplify(u1),
          "|  u_2 matches FH(3.5):", sp.simplify(u2-fh) == 0)
```

2.3 The $N = 2$ anchor

Convention C4 — Virasoro free-field construction verbatim [FH25] (2.8), (2.9)

$$T =: JJ : +\alpha\,\partial J, \qquad L_m = \sum_n : J_{m-n} J_n : -(m+1)\alpha J_m, \qquad c = 1 - 6\alpha^2.$$

Here $: \cdot \, :$ is standard mode normal ordering (positive modes to the right).

Consistency check 2.2 derived

Setting $N = 2$ in C3 with $J = J_2 = -J_1$: $u_2 = -J_1 J_2 + \alpha \partial J_2 = J^2 + \alpha \partial J$, matching C4. For the central charge, C3 gives $c^{\text{cl}} = -(1)(2)(3)\alpha^2 = -6\alpha^2$, and C4 gives $c = 1 - 6\alpha^2$: the difference is exactly the quantum $+1$ from the free boson, as it must be. Both are consistent.

2.4 Potentials, and the sign of the logarithm

Screening operators are exponentials of the potential, not of the current, so we must fix the potential's two-point function. This is the single most dangerous convention in the entire project: the sign here propagates into the screening condition, and a slip would flip F_0.

Convention trap W1 — the logarithm sign is not free

Define φ_i by $J_i = \partial \varphi_i$ and postulate $\varphi_i(z)\varphi_j(w) \sim A_{ij} \log(z - w)$. Then, since $\partial_w \log(z - w) = -(z - w)^{-1}$ and $\partial_z \partial_w \log(z - w) = +(z - w)^{-2}$,

$$J_i(z) J_j(w) = \partial_z \partial_w \big[\varphi_i(z)\varphi_j(w)\big] \sim \frac{A_{ij}}{(z - w)^2}.$$

Matching to C1, $J_i(z) J_j(w) \sim \kappa_{ij} (z - w)^{-2}$, forces

$$\boxed{\varphi_i(z)\varphi_j(w) \sim +\kappa_{ij} \log(z - w)} \qquad \text{(note the + sign)}$$

The plus sign is a direct consequence of [FH25]'s choice $J(z)J(w) \sim +\frac{1}{2}(z-w)^{-2}$ (their footnote 3), which is itself opposite to the more common $-$ convention of much of the Coulomb-gas literature. In the common convention one writes $\varphi\varphi \sim -\log$ and inserts factors of i into vertex operators. **This is a field redefinition, and it must be exhibited, not hidden.** Let ϕ_i be a canonically normalised real boson with the standard sign $\phi_i(z)\phi_j(w) \sim -\kappa_{ij} \log(z - w)$. Then

$$\varphi_i := i\,\phi_i \qquad \Longrightarrow \qquad \varphi_i\varphi_j \sim i^2(-\kappa_{ij}\log) = +\kappa_{ij}\log(z-w), \qquad J_i = \partial\varphi_i = i\,\partial\phi_i.$$

So [FH25]'s current normalisation is the statement $J = i\partial\phi$ in terms of a real boson — consistent with the explicit factor of i in their cylinder mode expansions $J_j(\theta) = i\sum_n J_{j,n}e^{in\theta}$, $\bar{J}_j(\theta) = i\sum_n \bar{J}_{j,n}e^{-in\theta}$ (3.11)–(3.12). **Three consequences must be carried everywhere:**

- A vertex operator with real momentum in our convention, $:e^{\beta\alpha_i\cdot\varphi}:$, is a vertex operator with imaginary momentum in the real-boson convention, $:e^{i\beta\alpha_i\cdot\phi}:$. The factor of i has not disappeared; it has been absorbed into the field.
- Hermiticity and reality conditions must therefore be stated in terms of ϕ, not φ. φ is anti-Hermitian in the real form; $:e^{\beta\varphi}:$ is unitary, not positive.
- Every screening momentum, zero mode and cocycle below is transported through $\varphi = i\phi$. Claims of "verbatim" adoption apply to [FH25]'s current normalisation $J_iJ_j \sim \kappa_{ij}(z-w)^{-2}$; the potential is introduced here, and the redefinition above is the bridge.

We nonetheless work with φ rather than ϕ, because it makes every current formula directly comparable with [FH25]. It does **not** make the vertex operators comparable with the standard Coulomb-gas literature without the translation above: in [FH25]'s own potential the screening momentum is imaginary. The price of the choice is exactly the bookkeeping listed above, and it must be paid explicitly in the paper's conventions section.

Definition D1 — vertex operator

For a constant covector $\lambda = (\lambda_i)$ we write $V_\lambda(z) :=: e^{\lambda\cdot\varphi(z)}:$. The basic contraction, derived from W1, is

$$J_i(z)\,V_\lambda(w) \;\sim\; \frac{(\kappa\lambda)_i}{z-w}\,V_\lambda(w), \qquad (\kappa\lambda)_i = \sum_j \kappa_{ij}\lambda_j.$$

3 The parent screening operators — the missing bridge

Why this section exists

[FH25] does not contain screening operators. The entire project consists of contracting objects that its authors never wrote down. Therefore the parent screenings must be imported from the Coulomb-gas literature[FL88],[BS93] and translated into [FH25]'s normalisation. Until that translation is done, symbols like b or β are meaningless and any statement of the form "F_0 fixes β" is vacuous. We do the translation from first principles rather than quoting a formula, because the sign convention of W1 is nonstandard and quoted formulas would not apply.

3.1 The weight computation, done explicitly

We work first at $N = 2$ in the single-field language of C2/C4, where $J = J_2$, $\varphi = \varphi_2$, and by W1 with $\kappa_{22} = \frac{1}{2}$,

$$\varphi(z)\varphi(w) \sim \tfrac{1}{2}\log(z-w), \qquad J(z)\varphi(w) \sim \frac{1/2}{z-w}.$$

Derivation 3.1 — conformal weight of $V_\beta =: e^{\beta\varphi}:$ derived

With $T =: JJ: + \alpha\partial J$ (C4), collect the three contributions to the singular part of $T(z)V_\beta(w)$.

(i) Double contraction of $:JJ:$. Each J contracts with the exponent, giving $\left(\frac{\beta}{2}\right)^2(z-w)^{-2}V_\beta = \frac{\beta^2}{4}(z-w)^{-2}V_\beta$.

(ii) Single contraction of $:JJ:$. Two ways, each giving $\frac{\beta}{2}(z-w)^{-1}:J(z)V_\beta(w):$; total $\beta(z-w)^{-1}:JV_\beta:$. Since $\partial_w V_\beta = \beta:JV_\beta:$, this is exactly $(z-w)^{-1}\partial_w V_\beta$ — the correct descendant term, which confirms the sign of W1. Had we chosen $\varphi\varphi \sim -\frac{1}{2}\log$, this term would have come out as $-\partial V_\beta$, an immediate contradiction. The descendant term is the sign-checker.

(iii) Background charge. $\alpha\,\partial_z J(z)V_\beta(w) \sim \alpha\beta\,\partial_z[\frac{1/2}{z-w}]V_\beta = -\frac{\alpha\beta}{2}(z-w)^{-2}V_\beta$.

Hence

$$T(z)V_\beta(w) \sim \frac{h_\beta}{(z-w)^2}V_\beta(w) + \frac{\partial_w V_\beta(w)}{z-w}, \qquad \boxed{h_\beta = \frac{\beta^2}{4} - \frac{\alpha\beta}{2}}$$

Definition D2 — screening condition

$Q = \oint dz\, V_\beta(z)$ commutes with all modes of T iff V_β has weight one, i.e. the singular part of TV_β is a total derivative. Therefore

$$h_\beta = 1 \quad\Longleftrightarrow\quad \beta^2 - 2\alpha\beta - 4 = 0. \qquad (\mathrm{F}^{\mathrm{rel}})$$

3.2 The two branches, and where $b + 1/b$ lives

Derivation 3.2 — branch structure derived

The quadratic F^{rel} has two roots $\beta_\pm = \alpha \pm \sqrt{\alpha^2+4}$, obeying

$$\beta_+\beta_- = -4, \qquad \beta_+ + \beta_- = 2\alpha.$$

Parametrising $\beta_+ = 2b$ and $\beta_- = -2/b$ — which is consistent, since $(2b)(-2/b) = -4$ automatically — the sum relation gives

$$\alpha = b - \frac{1}{b}, \qquad \text{and hence} \qquad c = 1 - 6\alpha^2 = 1 - 6\Big(b - \frac{1}{b}\Big)^2,$$

which matches the standard Coulomb-gas central charge after the real-form continuation of W1 (see the qualification in the intuition box below); it is a convention match, not an identity with every Liouville normalisation.[DF84],[BS93] This is the **validation of the whole normalisation dictionary**: an independent, well-known relation is reproduced with no free parameters. Our conventions are the standard ones in disguise.

Notation reserved from here on. The symbol b denotes the Toda coupling and nothing else, related to the single-field screening momentum by $\beta_+ = 2b$. The symbol β (with subscripts β_1, β_3 after contraction) denotes the full single-field momentum, i.e. the root of F^{rel}. The two differ by a factor of two and must never be substituted for one another: $\beta^2 - 2\alpha\beta - 4 = 0$ but $b^2 - \alpha b - 1 = 0$.

Physical intuition — what the two branches are

A Coulomb-gas boson with a background charge is a free field whose stress tensor has been "tilted" so that the vacuum carries charge. The tilt makes exactly two vertex operators marginal: the pair $(b, -1/b)$, dual under $b \to -1/b$. Physically they are the two ways to add a unit of screening charge without disturbing conformal symmetry. The familiar Liouville statement — two exponential potentials with $Q = b + b^{-1}$, the second invisible semiclassically — is the same structure in a different real form: our locked convention gives $c = 1 - 6(b - b^{-1})^2$ and $(\beta_+, \beta_-) = (2b, -2/b)$, with a relative minus sign and $b - b^{-1}$ in place of $b + b^{-1}$, reflecting the $\varphi = i\phi$ continuation of W1. We therefore state this as a Coulomb-gas dual-screening pair rather than importing Liouville formulas, which would require the explicit continuation to be written out. The reciprocity $\beta_+\beta_- = -4$ is the "$b + 1/b$" structure. Remember this: in §7 we will see the Carrollian contraction keep one branch and blow up the other — and the blown-up branch does not vanish from the physics, it becomes the background charge.

pythonCell 2 — verify the parent weight, the branch relations, and c = 1 − 6(b − 1/b)²

```
import sympy as sp
beta, alpha = sp.symbols('beta alpha')
bsym = sp.Symbol('b', nonzero=True)

h = beta**2/4 - alpha*beta/2                    # Derivation 3.1
Frel = sp.expand(4*(h - 1))                     # h = 1  ->  beta^2 - 2 alpha beta - 4 = 0
print("F_rel :", sp.factor(Frel), "= 0")

roots = sp.solve(sp.Eq(h,1), beta)
print("roots :", [sp.simplify(r) for r in roots])
```

```
print("product of roots (must be -4):", sp.simplify(sp.expand(roots[0]*roots[1])))
print("sum     of roots (must be 2a):", sp.simplify(sp.expand(roots[0]+roots[1])))

# parametrisation beta_+ = 2b, beta_- = -2/b
chk_prod = sp.simplify((2*bsym)*(-2/bsym) + 4)
alpha_of_b = sp.simplify(((2*bsym) + (-2/bsym))/2)
c_of_b = sp.simplify(1 - 6*alpha_of_b**2)
print("2b * (-2/b) + 4 =", chk_prod, " (0 means consistent)")
print("alpha(b)        =", alpha_of_b)
print("c(b)            =", sp.simplify(sp.expand(c_of_b)),
      " == 1-6(b-1/b)^2 ?", sp.simplify(c_of_b - (1 - 6*(bsym - 1/bsym)**2)) == 0)
```

3.3 The A_{N-1} case and the root dictionary

Convention C5 — roots imported [FL88],[BS93]

The simple roots of A_{N-1} in the N-component basis are $\alpha_i = e_i - e_{i+1}$, $i = 1, \dots, N-1$. Since $\sum_a (\alpha_i)_a = 0$, the trace-subtraction in κ is invisible on roots:

$$(\alpha_i, \alpha_j)_\kappa := \alpha_i^a \kappa_{ab} \alpha_j^b = \alpha_i \cdot \alpha_j = \begin{cases} 2 & i = j \\ -1 & |i-j| = 1 \\ 0 & \text{otherwise,} \end{cases}$$

i.e. the A_{N-1} Cartan matrix. The parent screening currents are $V_i =: e^{\mu\, \alpha_i \cdot \varphi} :$, with μ the root-basis momentum.

Proposition 3.3 — general-root screening weight derived

In the locked normalisation, $V_{\mu\alpha_i}$ has u_2-weight

$$\boxed{h(\mu\alpha_i) = \mu^2 + \alpha\mu}, \qquad \text{so the screening condition is} \qquad \mu^2 + \alpha\mu - 1 = 0.$$

Proof. Three steps, each elementary but each a place where a factor can be lost.

(i) Rewrite the quadratic part. Squaring the traceless constraint $\sum_a J_a = 0$ gives $\sum_a J_a^2 + 2\sum_{a<b} J_a J_b = 0$, hence $-\sum_{a<b} J_a J_b = \frac{1}{2}\sum_a J_a^2$, and C3 becomes $T = \frac{1}{2}\sum_a J_a^2 + \alpha \sum_{j>1}(j-1)\partial J_j$.

(ii) Quadratic contribution. Since $\lambda = \mu\alpha_i$ is traceless, $\kappa\lambda = \lambda$, so by D1 $J_a(z) V_\lambda(w) \sim \lambda_a (z-w)^{-1} V_\lambda$. Double contraction of $\frac{1}{2}\sum_a J_a^2$ gives $\frac{1}{2}\lambda^2 = \frac{1}{2}\mu^2(\alpha_i, \alpha_i) = \mu^2$.

(iii) Improvement contribution. One contraction gives $-\alpha \sum_{j>1}(j-1)\lambda_j$. For $\alpha_i = e_i - e_{i+1}$ one has $\sum_{j>1}(j-1)(\alpha_i)_j = -1$ for every i: for $i = 1$ only $j = 2$ survives, giving -1; for $i \geq 2$ one gets $(i-1) - i = -1$. Hence the contribution is $+\alpha\mu$. ■

Consistency with §3.1. At $N = 2$, W2 gives $\beta^{\text{sf}} = -2\mu$, and $\mu^2 + \alpha\mu = \frac{(\beta^{\text{sf}})^2}{4} - \frac{\alpha\beta^{\text{sf}}}{2}$ — exactly Derivation 3.1. The single-field and root computations are one computation. Cell 2b verifies this.

pythonCell 2b — verify Proposition 3.3: general-root weight and its N=2 reduction

```
import sympy as sp
mu, alpha = sp.symbols('mu alpha')

def root_weight(n, i):
    """h(mu*alpha_i) for A_{n-1}, alpha_i = e_i - e_{i+1}, 1-indexed i."""
    lam = [0]*n
    lam[i-1] += mu; lam[i] -= mu                # lambda = mu*(e_i - e_{i+1})
    quad  = sp.Rational(1,2)*sum(l**2 for l in lam)          # (1/2) lambda^2
    impr  = -alpha*sum((j)*lam[j] for j in range(1, n))      # -alpha sum_{j>1}(j-1)lambda_j
    return sp.expand(quad + impr)

for n in (2,3,4,5):
    for i in range(1, n):
        h = root_weight(n, i)
        assert sp.simplify(h - (mu**2 + alpha*mu)) == 0, (n,i,h)
    print("A_%d : h(mu alpha_i) = mu^2 + alpha*mu for all simple roots" % (n-1))

# N=2 reduction to the single-field convention, beta_sf = -2 mu
beta_sf = sp.Symbol('beta_sf')
h2 = root_weight(2,1).subs(mu, -beta_sf/2)
print("\nN=2 :", sp.expand(h2), " == beta^2/4 - alpha*beta/2 ?",
      sp.simplify(h2 - (beta_sf**2/4 - alpha*beta_sf/2)) == 0)
print("screening condition x4 :", sp.expand(4*(h2-1)), "= 0")

# root-basis F0 by contraction
e, ap, am, mu1, mu3 = sp.symbols('epsilon alpha^+ alpha^- mu_1 mu_3')
```

```
F = (e*mu1+e**3*mu3)**2 + (ap/e+e*am)*(e*mu1+e**3*mu3) - 1
print("\nroot-basis F0 :", sp.expand(sp.series(F,e,0,3).removeO().coeff(e,0)), "= 0")
```

Proposition 3.4 — the parent screening identity, derived derived

Proposition 3.3 fixed the u_2 weight. The remaining question — whether the same μ screens the higher currents u_k — was previously imported from Fateev–Lukyanov. **It is now derived directly in our conventions, and the import is eliminated.**

Statement — a computer-assisted proposition over a finite range, not a general-N theorem. For A_{N-1} with $N \le 4$, every simple root α_i, and every $k = 2, \dots, N$,

$$\boxed{\mathrm{Res}_{w=z}\Big[u_k(z)\,V_{\mu\alpha_i}(w)\Big] \;=\; (\mu^2+\alpha\mu-1)\;\mathcal{R}_{k,i}(z)\,V_{\mu\alpha_i}(z)}$$

where $\mathcal{R}_{k,i}$ is a **differential** polynomial in the currents — for the second A_2 simple root the u_3 quotient contains ∂J terms, so "polynomial in the currents" (an earlier wording) is too narrow.

with $|z| > |w|$, in the single field order fixed by C3q. **The vertex factor is part of the identity.** The symbolic routine computes the coefficient after factoring V out, so $\mathcal{R}_{k,i}$ is a differential polynomial in the currents and the displayed operator identity carries $V_{\mu\alpha_i}(z)$ explicitly; an earlier draft omitted it, which contradicted this notebook's own $k = 2$ sanity check, where the residue is $(h-1)\,\partial V$.

for a differential polynomial $\mathcal{R}_{k,i}$ in the currents. Hence $[Q_i, u_k] = 0$ precisely on the screening locus of Proposition 3.3 — the same locus for every k, which is exactly the content of the theorem previously imported from [FL88].

The locus, stated exactly. Every residue above carries an overall factor μ, so Res $= 0$ holds on the union of two branches:

$$\big\{\,\mu^2+\alpha\mu = 1\,\big\} \;\cup\; \big\{\,\mu = 0\,\big\}.$$

The second is trivial and must be excluded: at $\mu = 0$ the density degenerates to the constant $V_0 = 1$ and the contour charge vanishes identically, so it screens nothing. **The screening locus is therefore** $\mu^2 + \alpha\mu = 1$ **with** $\mu \neq 0$, and the word "precisely" in the statement above is to be read with that exclusion.

How the divisibility is tested (this matters). A previous revision tested divisibility by setting $q := R_k/F$ and checking $R_k - Fq = 0$ — a tautology, true for every R_k. That test established nothing. The present test checks two independent conditions: that cancel(R_k/F) has denominator one, i.e. the quotient is polynomial; and that R_k vanishes identically under $\alpha \to (1-\mu^2)/\mu$, i.e. on the locus itself. A previous revision also differentiated the vertex with an independent J_N after eliminating it; the derivation now uses $\lambda \cdot J = \sum_{a<N}(\lambda_a - \lambda_N)J_a$. The second defect was invisible for $\alpha_1 = e_1 - e_2$ (where $\lambda_N = 0$) but did affect the other roots.

Convention C3q — quantum Miura ordering and residue orientation

Proposition 3.4 is a quantum OPE statement, whereas C3 introduced the Miura operator classically. Two things must therefore be fixed before the proposition means anything.

(i) Ordering. The currents u_k are defined by the same product as C3 with the standard global normal ordering on the plane (positive modes to the right), applied to the whole coefficient of each $(\alpha\partial)^{N-k}$; this is [FH25]'s quantum Miura transformation, their §4.1 with (4.1). Since the only contractions occurring below are between the J's of u_k and the exponent of V — the J's inside u_k are normal-ordered among themselves and do not self-contract — the leftover fields may be treated as commuting symbols when extracting the singular coefficients.

(ii) Orientation. All residues are written in the fixed order and region

$$\mathrm{Res}_{w=z}\,\big[\,u_k(z)\,V(w)\,\big], \qquad |z| > |w|,$$

with the OPE expanded in $(z-w)$ and the contour taken counterclockwise in w about $w = z$. The criterion $[Q_i, u_k] = 0$ is the vanishing of this residue.

Method. The criterion is the vanishing of the residue, $\mathrm{Res}_{w=z} = \sum_{p\geq 1} \frac{(-1)^p}{(p-1)!}\mathcal{D}^{p-1}\mathcal{O}_p$, where $\mathrm{Sing}\big[u_k(z)V(w)\big] = \sum_p \mathcal{O}_p(w)V(w)(z-w)^{-p}$ and $\mathcal{D}\mathcal{O} = \partial\mathcal{O} + \mathcal{O}\,(\lambda\cdot J)$ is the derivation that also differentiates V. The $\mathcal{O}_p$ are obtained by Wick contraction: only J–V pairings occur (the J's inside u_k are already normal-ordered among themselves), with $\partial^m J_a(z)V(w) \sim \lambda_a(-1)^m m!\,(z-w)^{-m-1}V(w)$, and the uncontracted fields Taylor-expanded about w. Sanity check on $k=2$: the criterion returns $(h-1)\partial V$, reproducing Definition D2.

Explicit results at $N=3$. For $\alpha_1 = e_1 - e_2$,

$$\mathrm{Res}\,\big[u_2 V\big] = \mu\,(J_1 - J_2)\,F\,V, \qquad \mathrm{Res}\,\big[u_3 V\big] = \mu\,(J_1 - J_2)(J_1 + J_2)\,F\,V, \qquad F := \mu^2 + \alpha\mu - 1,$$

with $J_1 - J_2 = \alpha_1\cdot J$ and $J_1 + J_2 = -J_3$; and for $\alpha_2 = e_2 - e_3$, in the reduced variables,

$$\mathrm{Res}\,\big[u_2 V\big] = \mu\,(J_1 + 2J_2)\,F\,V.$$

Cell 2c performs the computation and applies the genuine divisibility test for $N = 2, 3, 4$, all simple roots, all u_k.

Consequence for the programme, stated narrowly. Paper 1 is the $N=3$ case, so it is **self-contained with respect to the parent screening identity**: Propositions 3.3 and 3.4 supply it in the locked conventions, and [FL88]/[BS93] are cited for the origin of the general-N result and for context rather than as load-bearing inputs. This is not the claim that the project has "no external dynamical input": [FH25] remains load-bearing for the symmetric Carrollian W_3 algebra, [DDF26] for the bulk charges, and standard free-field/intertwiner machinery throughout.

Total-derivative witnesses at $N=3$ derived

Residue-vanishing and total-derivative form are equivalent, but the witness is what makes the result hand-auditable rather than a black-box divisibility output, so it is displayed here rather than deferred. Writing $\mathrm{Sing}\big[u_k(z)V(w)\big] = \sum_p \mathcal{O}_p(w)V(w)(z-w)^{-p}$ and solving $\mathcal{O}_q = \mathcal{D}\mathcal{B}_q + (q-1)\mathcal{B}_{q-1}$ top-down on the locus $\alpha = (1-\mu^2)/\mu$, every residual $\mathcal{O}_1 - \mathcal{D}\mathcal{B}_1$ vanishes identically and the witnesses are:

$$\mathrm{Sing}\big[u_2(z)V_i(w)\big] = \partial_w\left(\frac{V_i(w)}{z-w}\right) \qquad \text{for both simple roots,}$$

$$\alpha_1: \quad \mathrm{Sing}\big[u_3(z)V_1(w)\big] = \partial_w\left(\frac{(J_1+J_2)\,V_1(w)}{z-w}\right),$$

$$\alpha_2: \quad \mathrm{Sing}\big[u_3(z)V_2(w)\big] = \partial_w\left(-\frac{\alpha\,V_2(w)}{(z-w)^2} - \frac{J_1\,V_2(w)}{z-w}\right).$$

The spin-two witness is the familiar one: $\mathcal{O}_2 = h = \mu^2 + \alpha\mu$ equals 1 on the locus, and $\mathcal{O}_1 = \lambda\cdot J = \partial V/V$. For α_2 at $k=3$ the computation returns $\mathcal{B}_2 = \mu - \mu^{-1}$, which is exactly $-\alpha$ on the locus — a small internal consistency check worth noting. Cell 2d computes all four witnesses and verifies that the residuals vanish; the verifier repeats the check.

Scope. This is the parent relativistic statement. It says nothing about the Carrollian contraction, averaged ordering, or the existence of limiting operators; those remain the content of Theorems 3–4.

pythonCell 2c — Proposition 3.4: direct residue computation, N = 2,3,4, all simple roots

```python
import sympy as sp
from itertools import combinations
al, mu, xx = sp.symbols('alpha mu x')

def miura_u(n):
    z = sp.Symbol('z'); Js = [sp.Function('J%d'%(a+1))(z) for a in range(n)]; f = sp.Function('f')(z)
    e = f
    for a in reversed(range(n)): e = al*sp.diff(e, z) - Js[a]*e
    e = sp.expand(e); us = {}; rem = sp.expand(e - al**n*sp.diff(f, z, n))
    for k in range(1, n+1):
        dk = sp.diff(f, z, n-k)
        co = sp.expand(rem.coeff(dk)).subs({sp.diff(f, z, j): 0 for j in range(1, n+1)})
        us[k] = sp.expand(sp.simplify(-co/al**(n-k))); rem = sp.expand(rem + us[k]*al**(n-k)*dk)
    return us, Js, z
```

```
def Pw(a, m): return sp.Symbol('P%d_%d' % (a, m))          # d^m J_a (w)

def to_terms(expr, Js, z, n, maxd=6):
    expr = sp.expand(expr); atoms = {}
    for a in range(n):
        for k in range(maxd): atoms[sp.diff(Js[a], z, k)] = (a, k)
    out = []
    for t in sp.Add.make_args(expr):
        c = sp.S(1); fl = []
        for fac in sp.Mul.make_args(t):
            b, e = fac.as_base_exp()
            if b in atoms: fl += [atoms[b]]*int(e)
            else: c *= fac
        out.append((c, fl))
    return out

def singular(terms, lam, maxpole):
    A = {p: sp.S(0) for p in range(1, maxpole+1)}
    for c, fl in terms:
        L = len(fl)
        for r in range(1, L+1):
            for S in combinations(range(L), r):
                pole = 0; coef = c
                for i in S:
                    a, k = fl[i]; coef *= lam[a]*(-1)**k*sp.factorial(k); pole += k+1
                rest = [fl[i] for i in range(L) if i not in S]
                ser = sp.S(1)
                for (a, k) in rest:
                    t = sum(Pw(a, k+j)*xx**j/sp.factorial(j) for j in range(pole))
                    ser = sp.expand(ser*t)
                    ser = sum(ser.coeff(xx, m)*xx**m for m in range(pole))
                for m in range(pole):
                    p = pole - m
                    if 1 <= p <= maxpole: A[p] += coef*sp.expand(ser).coeff(xx, m)
    return {p: sp.expand(v) for p, v in A.items()}

def Dder(expr, lam, n, maxm=11):
    # TRACE-AWARE.  J_N has been eliminated via sum_a J_a = 0, so the independent
    # fields are J_1..J_{N-1} and the vertex derivative must use
    #     lam . J = sum over a < N of (lam_a - lam_N) J_a ,
    # NOT sum over all a of lam_a J_a with an independent J_N.
    out = sp.S(0)
    for a in range(n-1):
        for m in range(maxm):
            d = sp.diff(expr, Pw(a, m))
            if d != 0: out += d*Pw(a, m+1)
    lam_dot_J = sum((lam[a] - lam[n-1])*Pw(a, 0) for a in range(n-1))
    return sp.expand(out + expr*lam_dot_J)

def residue(A, lam, n, maxpole):
    Rr = sp.S(0)
    for p in range(1, maxpole+1):
        t = A[p]
        for _ in range(p-1): t = Dder(t, lam, n)
        Rr += sp.Rational((-1)**p, sp.factorial(p-1))*t
    return sp.expand(Rr)

screen = mu**2 + al*mu - 1
for N in (2, 3, 4):
    us, Js, z = miura_u(N)
    sub = {sp.diff(Js[N-1], z, j): -sum(sp.diff(Js[a], z, j) for a in range(N-1)) for j in range(7)}
    for i in range(N-1):
        lam = [sp.S(0)]*N; lam[i] = mu; lam[i+1] = -mu
        res = []
        for k in range(2, N+1):
            expr = sp.expand(us[k].subs(sub)); mp = k+1
            A = singular(to_terms(expr, Js, z, N), lam, mp)
            Rk = sp.expand(residue(A, lam, N, mp))
            # GENUINE divisibility test.  NOTE: writing q = cancel(Rk/screen) and then
            # checking Rk - screen*q == 0 is a TAUTOLOGY (true for every Rk) and was the
            # defect in an earlier revision.  We test two independent things:
            #   (t1) the quotient is a polynomial  (denominator cancels to 1);
            #   (t2) Rk vanishes identically on the screening locus alpha = (1-mu^2)/mu.
            qq = sp.cancel(Rk/screen); num, den = sp.fraction(sp.together(qq))
            t1 = sp.simplify(den - 1) == 0
            t2 = sp.simplify(Rk.subs(al, (1-mu**2)/mu)) == 0
            t3 = sp.simplify(sp.expand(Rk - screen*sp.expand(num))) == 0
            res.append((k, t1 and t2 and t3))
        print('N=%d  root alpha_%d :' % (N, i+1),
              '  '.join('u_%d %s' % (k, 'PASS' if ok else 'FAIL') for k, ok in res))

# explicit N=3 residues
us, Js, z = miura_u(3)
sub = {sp.diff(Js[2], z, j): -(sp.diff(Js[0], z, j) + sp.diff(Js[1], z, j)) for j in range(7)}
lam = [mu, -mu, sp.S(0)]
print()
for k in (2, 3):
    A = singular(to_terms(sp.expand(us[k].subs(sub)), Js, z, 3), lam, k+1)
    print('Res[u_%d V] / V =' % k, sp.factor(sp.simplify(residue(A, lam, 3, k+1))))
```

pythonCell 2d — total-derivative witnesses at N=3, both simple roots

```
import sympy as sp
# reuses miura_u, to_terms, singular, Dder, Pw from Cell 2c
onF = {al: (1-mu**2)/mu}                       # the screening locus

def witnesses(N, root):
    us, Js, z = miura_u(N)
    sub = {sp.diff(Js[N-1], z, r): -sum(sp.diff(Js[a], z, r) for a in range(N-1)) for r in range(10)}
    lam = [sp.S(0)]*N; lam[root] = mu; lam[root+1] = -mu
    for k in range(2, N+1):
        A = singular(to_terms(sp.expand(us[k].subs(sub)), Js, z, N), lam, k+1)
        A = {p: sp.simplify(sp.expand(v.subs(onF))) for p, v in A.items()}
        Pmax = max(A); B = {p: sp.S(0) for p in range(0, Pmax+2)}
        for q in range(Pmax, 1, -1):
            B[q-1] = sp.simplify(sp.expand((A[q] - Dder(B[q], lam, N).subs(onF))/(q-1)))
        residual = sp.simplify(sp.expand(A[1] - Dder(B[1], lam, N).subs(onF)))
        wit = ", ".join("B_%d = %s" % (p, sp.simplify(B[p])) for p in range(1, Pmax+1) if B[p] != 0)
        print("  u_%d :  %s   |  residual O_1 - D(B_1) = %s" % (k, wit, residual))
        assert residual == 0, (N, root, k, residual)

for root, lbl in ((0, 'alpha_1'), (1, 'alpha_2')):
    print("N=3, root %s  (Sing = d_w[ sum_p B_p V /(z-w)^p ])" % lbl)
    witnesses(3, root)
print("\nAll residuals vanish: the singular parts ARE total derivatives on the locus.")
print("Note B_2 = mu - 1/mu for alpha_2, k=3, which equals -alpha on F=0.")
```

Convention trap W2 — two incompatible meanings of β

At $N = 2$ the constraint gives $\varphi_1 = -\varphi_2$, so $\alpha_1 \cdot \varphi = \varphi_1 - \varphi_2 = -2\varphi$ with $\varphi := \varphi_2$. Hence the root-language momentum β^{root} and the single-field momentum β^{sf} of §3.1 are related by

$$\beta^{\mathrm{sf}} = -2\,\beta^{\mathrm{root}}.$$

Consistency check on the propagator: $\langle(\alpha_1 \cdot \varphi)^2\rangle = (\alpha_1, \alpha_1)_\kappa \log = 2\log$, and directly $\langle(-2\varphi)^2\rangle = 4\kappa_{22}\log = 2\log$. [checked] **Every numerical statement below carries an explicit superscript.** Confusing the two is the single most likely arithmetic error in this project.

4 The Carrollian contraction

4.1 Carroll-compatible combinations

Convention C6 verbatim [FH25] (3.7)–(3.10)

$$U_k^+ = \frac{\epsilon^k}{2}(u_k + \bar{u}_k), \qquad U_k^- = \frac{\epsilon^{k-2}}{2}(u_k - \bar{u}_k),$$

$$J_i^+ = \frac{\epsilon}{2}(J_i + \bar{J}_i), \qquad J_i^- = \frac{\epsilon^{-1}}{2}(J_i - \bar{J}_i), \qquad \alpha^+ = \frac{\epsilon}{2}(\alpha + \bar{\alpha}), \qquad \alpha^- = \frac{\epsilon^{-1}}{2}(\alpha - \bar{\alpha}),$$

with the inverse relations

$$u_k = \epsilon^{-k}U_k^+ + \epsilon^{2-k}U_k^-, \qquad \bar{u}_k = \epsilon^{-k}U_k^+ - \epsilon^{2-k}U_k^-,$$

$$J_i = \epsilon^{-1}J_i^+ + \epsilon J_i^-, \qquad \bar{J}_i = \epsilon^{-1}J_i^+ - \epsilon J_i^-, \qquad \alpha = \epsilon^{-1}\alpha^+ + \epsilon\alpha^-, \qquad \bar{\alpha} = \epsilon^{-1}\alpha^+ - \epsilon\alpha^-.$$

Convention C7 — modes and contracted brackets verbatim [FH25] (3.13), (3.15)

$$J_{i,n}^+ = \frac{\epsilon}{2}(J_{i,n} + \bar{J}_{i,-n}), \qquad J_{i,n}^- = \frac{\epsilon^{-1}}{2}(J_{i,n} - \bar{J}_{i,-n}),$$

$$[J_m^{(i)+}, J_n^{(j)+}]^{\mathrm{cl}} = 0, \qquad [J_m^{(i)-}, J_n^{(j)-}]^{\mathrm{cl}} = 0, \qquad [J_m^{(i)-}, J_n^{(j)+}]^{\mathrm{cl}} = \frac{m}{2}\,\delta_{m,-n}\,\kappa^{ij}.$$

Note the mode reversal $\bar{J}_{i,-n}$ in the barred sector: this is the ultrarelativistic contraction as opposed to the Galilean one, and it is the origin of the averaged ordering in §5.

4.2 A structural fact: the Heisenberg sector does not contract

Observation 4.1 derived — ϵ-exactness of the oscillator brackets

The three brackets of C7 hold identically for every $\epsilon \neq 0$, not merely in the limit $\epsilon \to 0$. The Heisenberg algebra of the Carroll-compatible modes is ϵ-independent; only the composite currents $U_k^{\pm}$ and the background charges α^+, α^- carry ϵ.

This is an immediate algebraic consequence of [FH25] (3.13)–(3.15) rather than a new theorem; it is recorded because it is used repeatedly and because its limits are easy to overstate.

Proof. Insert C7 into C1, using $[J, \bar{J}] = 0$ (independent copies) and $[\bar{J}_{i,m}, \bar{J}_{j,n}] = m\kappa_{ij}\delta_{m,-n}$:

$$[J^+_{i,m}, J^+_{j,n}] = \frac{\epsilon^2}{4}\Big([J_{i,m}, J_{j,n}] + [\bar{J}_{i,-m}, \bar{J}_{j,-n}]\Big) = \frac{\epsilon^2}{4}\Big(m\kappa_{ij}\delta_{m,-n} + (-m)\kappa_{ij}\delta_{m,-n}\Big) = 0,$$

where the crucial cancellation is the sign flip $(-m)$ produced by the mode reversal. Identically, $[J^-, J^-] = \frac{\epsilon^{-2}}{4}(m - m)\kappa\delta = 0$. For the crossed bracket the ϵ-factors cancel between the two prefactors:

$$[J^-_{i,m}, J^+_{j,n}] = \frac{\epsilon^{-1}}{2} \cdot \frac{\epsilon}{2}\Big(m\kappa_{ij}\delta_{m,-n} - (-m)\kappa_{ij}\delta_{m,-n}\Big) = \frac{m}{2}\kappa_{ij}\delta_{m,-n}. \qquad \blacksquare$$

What this does and does not give. It gives one bounded algebraic fact: a common formal oscillator algebra may be chosen once and for all, independent of ϵ, so that the ϵ-dependence of the construction sits entirely in the composite currents $U_k^{\pm}$ and the background charges α^+, α^-. That is genuinely convenient, and it is all that is claimed.

It does not give representation rigidity, and the earlier draft of this notebook wrongly said it did. The map $(J, \bar{J}) \mapsto (J^+, J^-)$ of C7 is a singular ϵ-dependent linear transformation: it need not be implementable by a unitary, the parent Fock vacua need not map to a common vacuum, the normal-ordering prescription is manifestly not preserved (that is the entire content of §5.1–5.2), and the zero-mode sector can and does change. Whether a common module exists on which the limiting operators act is a separate construction, begun in §6 and completed in Notebook II. No argument in §7 may lean on "rigidity".

Falsifiable content. If any of the three brackets acquired ϵ-dependence, the observation is false. Cell 3 verifies all three on the delta support.

Check 4.2 — symmetry of the crossed pairing derived

[FH25] states the crossed bracket in the order $[J^-, J^+]$ in (3.15) but in the order $[J^+, J^-]$ in (2.11). These are consistent, and not by accident: from $[J^-_m, J^+_n] = \frac{m}{2}\kappa\delta_{m,-n}$ and antisymmetry, $[J^+_m, J^-_n] = -\frac{n}{2}\kappa\delta_{m,-n} = +\frac{m}{2}\kappa\delta_{m,-n}$, because $n = -m$ on the support of the delta. **The crossed pairing is symmetric under exchange of the $\pm$ labels.** At $N = 2$, $\kappa_{22} = \frac{1}{2}$ gives $[J^+_m, J^-_n] = \frac{m}{4}\delta_{m,-n}$, exactly [FH25] (2.11). [checked]

pythonCell 3 — Observation 4.1 on the δ-support (ε-exactness); N=2 specialisation m/4

```
import sympy as sp
m, n, e = sp.symbols('m n epsilon', nonzero=True)
kap = sp.Symbol('kappa_ij')

# parent brackets (FH 3.14) as functions of mode labels; delta support enforced by n = -m
def par(mm, nn):            # [J_{i,mm}, J_{j,nn}] with delta_{mm,-nn} imposed
    return mm*kap if sp.simplify(mm + nn) == 0 else 0

# C7 with n = -m so the deltas are saturated
mp, mn = m, -m
# [J+_m, J+_n]  = (e^2/4) ( [J_m,J_n] + [Jbar_{-m}, Jbar_{-n}] )
JpJp = sp.simplify( e**2/4 * ( par(mp,mn) + par(-mp,-mn) ) )
# [J-_m, J-_n]  = (e^-2/4) ( [J_m,J_n] + [Jbar_{-m}, Jbar_{-n}] )   (signs: (-)(-) = +)
JmJm = sp.simplify( e**-2/4 * ( par(mp,mn) + par(-mp,-mn) ) )
# [J-_m, J+_n]  = (e^-1/2)(e/2) ( [J_m,J_n] - [Jbar_{-m}, Jbar_{-n}] )
JmJp = sp.simplify( sp.Rational(1,4) * ( par(mp,mn) - par(-mp,-mn) ) )

print("[J+_m , J+_-m] =", JpJp, "     (must be 0, for ALL epsilon)")
print("[J-_m , J-_-m] =", JmJm, "     (must be 0, for ALL epsilon)")
print("[J-_m , J+_-m] =", JmJp, "     (must be m*kappa/2, epsilon-free)")
print()
print("epsilon-free?  ", all(sp.simplify(sp.diff(x, e)) == 0 for x in (JpJp, JmJm, JmJp)))
# LIMITATION: n = -m is imposed from the start, so this checks the coefficient on the
# nonzero Kronecker-delta support, not the full distributional commutator.
```

```
print("N=2 value (kappa_22 = 1/2):", sp.simplify(JmJp.subs(kap, sp.Rational(1,2))),
      " -> matches FH (2.11) m/4:", sp.simplify(JmJp.subs(kap, sp.Rational(1,2)) - m/4) == 0)
```

4.3 The contracted Miura transformation

Convention C8 — Carrollian Miura transformation verbatim [FH25] (1.1)–(1.2), = (3.23)/(3.26)

$$(\alpha^+\partial - J_1^+)\cdots(\alpha^+\partial - J_N^+) = (\alpha^+\partial)^N - \sum_{k=2}^{N} U_k^+(\alpha^+\partial)^{N-k},$$

$$\sum_{j=1}^{N}(\alpha^+\partial - J_1^+)\cdots J_j^-\cdots(\alpha^+\partial - J_N^+) = \sum_{k=2}^{N}\Big(U_k^- - \alpha^-\frac{\partial}{\partial\alpha^+}U_k^+\Big)(\alpha^+\partial)^{N-k},$$

where in the second line the j-th factor is replaced by J_j^-, and $\sum_i J_i^\pm = 0$ (3.27).

Physical intuition — a differential operator and its first jet

C8 says something clean that is easy to lose in the indices. The $+$ sector is an ordinary Miura factorisation with background charge α^+: it defines the currents U_k^+ exactly as in the relativistic case, but built from commuting fields. The $-$ sector is the derivative of that structure: the left-hand side is the first-order variation of the product under $J_j^+ \to J_j^+ + \delta J_j^-$, and the right-hand side is correspondingly the variation of the invariants, corrected by $\alpha^-\,\partial_{\alpha^+}U_k^+$ because the background charge is varied too. The Carrollian W_N-algebra therefore has a Takiff-like first-jet structure over the abelian $+$-sector, with U^- playing the role of a fibre coordinate. (This is an organising interpretation; no Lie- or vertex-algebra isomorphism to a Takiff double is claimed or proved here.) Every structural fact below — the one-J^- filtration, the screening doublet, the Jordan-block pattern — is a shadow of this single statement.

Convention C9 — identification of T, M and central charges verbatim [FH25] (3.28)–(3.31)

$$T = \sum_n L_n e^{in\theta} = 2U_2^-, \qquad M = \sum_n M_n e^{in\theta} = 2U_2^+,$$

$$c_L^{\rm cl} = -4(N-1)N(N+1)\alpha^+\alpha^-, \qquad c_M^{\rm cl} = -2(N-1)N(N+1)(\alpha^+)^2.$$

4.4 The $N = 2$ currents, explicitly

Convention C10 verbatim [FH25] (3.32), (3.34)

With $J^\pm := J_2^\pm = -J_1^\pm$,

$$M = 2U_2^+ = 2\,J^+J^+ + 2\alpha^+\,\partial J^+, \qquad T = 2U_2^- = 4\,J^+J^- + 2\alpha^+\,\partial J^- + 2\alpha^-\,\partial J^+.$$

Check 4.3 — recovering C10 from the parent by contraction derived

[FH25] verify this in their (3.36)–(3.41): substituting $J = \epsilon^{-1}J^+ + \epsilon J^-$, $\bar J = \epsilon^{-1}J^+ - \epsilon J^-$ into $T = JJ + \alpha\partial J$, $\bar T = \bar J\bar J + \bar\alpha\partial\bar J$, and forming $\epsilon^2(T+\bar T)/1$ resp. $T - \bar T$. Cell 4 reproduces both, including the cancellation of the ϵ^{-2} and ϵ^{-1} divergences.

pythonCell 4 — CLASSICAL/commuting contraction of T, T̄ vs FH (3.32),(3.34); no ordering content

```
import sympy as sp
zz = sp.Symbol('z'); e = sp.Symbol('epsilon', positive=True)
Jp = sp.Function('Jp')(zz); Jm = sp.Function('Jm')(zz)
apS, amS = sp.symbols('alpha^+ alpha^-')

J    =  Jp/e + e*Jm                    # FH (3.10b)
Jbar =  Jp/e - e*Jm
al   =  apS/e + e*amS                  # FH (3.10c)
albar=  apS/e - e*amS

T    = J*J    + al*sp.diff(J, zz)      # FH (3.35)
Tbar = Jbar*Jbar + albar*sp.diff(Jbar, zz)

M_field = sp.expand(e**2*(T + Tbar))   # = 2 U^+_2   (FH 3.36)
T_field = sp.expand(T - Tbar)          # = 2 U^-_2   (FH 3.39)

M_lim = sp.simplify(sp.limit(M_field, e, 0))
T_lim = sp.simplify(sp.limit(T_field, e, 0))
```

```
M_target = 2*Jp**2 + 2*apS*sp.diff(Jp, zz)                                  # FH (3.32)
T_target = 4*Jp*Jm + 2*apS*sp.diff(Jm, zz) + 2*amS*sp.diff(Jp, zz)          # FH (3.34)

print("M limit :", sp.simplify(M_lim))
print("  == FH (3.32)?", sp.simplify(M_lim - M_target) == 0)
print("T limit :", sp.simplify(T_lim))
print("  == FH (3.34)?", sp.simplify(T_lim - T_target) == 0)
print()
# check that T - Tbar has NO negative powers of epsilon at all (exact cancellation)
ser    = sp.expand(T_field)
negpow = [t for t in ser.as_ordered_terms()
          if sp.Add(*[sp.S(ex) for bs, ex in t.as_powers_dict().items() if bs == e] or [sp.S(0)]) < 0]
print("terms of T-Tbar with negative powers of epsilon:",
      negpow if negpow else "none  (exact cancellation)")
```

5 The quantum layer: ordering, and the filtration lemma

5.1 Flipped versus symmetric

[FH25] distinguish two quantum contractions. In the flipped construction the time direction — and therefore the normal-ordering prescription — is reversed in one copy; the result is isomorphic to the quantum Galilean W_N-algebra, with central charges[FH25] (1.3)

$$\tilde{c}_L = (N-1)\big[2 - 4N(N+1)\alpha^+\alpha^-\big], \qquad \tilde{c}_M = -2(N-1)N(N+1)(\alpha^+)^2.$$

In the symmetric construction the time direction is kept in both copies; the price is that one must use an averaged ordering, and the reward is that the central charges coincide with the classical ones (1.4):

$$c_L = -4(N-1)N(N+1)\alpha^+\alpha^-, \qquad c_M = -2(N-1)N(N+1)(\alpha^+)^2.$$

Which algebra this project is about

The symmetric one. The flipped/Galilean algebra is a different object with different central charges and it already has free-field screening literature in the $W(2,2)$ guise[AR16]. Our novelty claim is specifically about the proper symmetric algebra with averaged ordering. Every statement in the paper must say ”symmetric” explicitly; a referee who reads ”Carrollian” as ”Galilean” will believe the result is known.

5.2 Averaged ordering

Convention C11 — averaged ordering verbatim [FH25] (2.14), (4.29)–(4.31)

$$:\!J^+_m J^-_{-m}\!: = \tfrac{1}{2}(J^+_m J^-_{-m} + J^-_{-m} J^+_m),$$

and for a general product containing exactly one J^-,

$$ \av{J^-_{j,n}J^+_{[\text{i.e. for one-}J^-\text{ products, averaged ordering is ”put the }J^-\text{ once at the far left and once at the far right, and average”. The rewriting is legitimate because }\big[J^+_{[(4.30).}$$

Convention C12 — the filtration verbatim [FH25], §4.4

”The fields U^+_j are built only from J^+_i, which all commute and do not require an ordering prescription. The fields U^-_j are built as sums of products of J^+ fields and at most a single J^- field.”

Physical intuition — why averaging removes the quantum corrections

Normal ordering is a choice of what ”positive frequency” means, i.e. a choice of time. The Carrollian contraction squeezes the light cone flat, so there is no longer a preferred time to make that choice with. Averaging normal and anti-normal ordering is the honest response: it is the symmetric prescription selected by the two-copy contraction — not proved to be the unique consistent one. [FH25]'s remarkable observation is that this symmetric prescription is closed — commutators of averaged one-J^- expressions come out already averaged (4.32)–(4.34) — so no reordering is needed and the classical structure constants survive quantisation untouched. A heuristic for why: in the Takiff-like picture of §4.3 there is no quadratic-in-fibre sector for quantum corrections to live in. **This is intuition only** — it is not a proof, and in particular it must not be used to pre-judge F_1 or the $U^\pm_3$ ordering question, both of which are open.

5.3 The filtration lemma, proved before any OPE

The following is the structural fact that makes the whole project a finite computation. It is proved here abstractly, from the commutation relations alone — not verified case by case on OPEs, which would prove nothing general.

Lemma 5.1 (Filtration / Truncation) formulated here

Let $\mathcal{F}^r$ denote the span of averaged-ordered products of the $J_i^{\pm}$ and their derivatives containing at most r factors of J^- (counting fields, not parameters). Let

$$\mathscr{S}^{(0)} =: e^{\beta\,\alpha_i\cdot\varphi^+}:, \qquad \mathscr{S}^{(1)} =: \big(A\,\alpha_i\cdot\varphi^- + B\,\alpha_i\cdot\varphi^+\big)e^{\beta\,\alpha_i\cdot\varphi^+}: .$$

Define the degree map $d(+)=0$, $d(-)=1$. Then, with $U_k^+ \in \mathcal{F}^0$ and $U_k^- \in \mathcal{F}^{\le 1}$ (C12), the number of Wick pairings contributing to $U_k^\sigma(z)\,\mathscr{S}^{(a)}(w)$ is bounded by $d(\sigma)+a$. The bound counts pairings; it is not a bound on the pole order, nor on the number of terms:

	$\mathscr{S}^{(0)}$	$\mathscr{S}^{(1)}$
U_k^+	0	1
U_k^-	1	2

Proof. The only nonvanishing pairing is the crossed one (C7), so a contraction requires one J^--type object and one J^+-type object. The exponent of $\mathscr{S}^{(a)}$ contains only φ^+; the prefactor of $\mathscr{S}^{(1)}$ contains at most one φ^- and at most one φ^+. Hence:

- $U_k^+\mathscr{S}^{(0)}$: both sides are pure $+$; $[\varphi^+,\varphi^+]=0$ gives zero contractions, so the OPE is **regular**.
- $U_k^-\mathscr{S}^{(0)}$: the unique J^- can pair with the exponent; nothing else can pair. One contraction.
- $U_k^+\mathscr{S}^{(1)}$: the J^+'s can pair only with the single φ^- in the prefactor. One contraction.
- $U_k^-\mathscr{S}^{(1)}$: the unique J^- pairs with exponent or with the φ^+ prefactor (one contraction), and one J^+ pairs with the φ^- prefactor (a second). No third pairing exists.

Moreover, because $[J^-,\varphi^+]$ is central, the Baker–Campbell–Hausdorff series terminates exactly:

$$\big[J^-_{a,m},\, :e^{\beta\alpha_i\cdot\varphi^+}:\big] = \beta\,\big[J^-_{a,m},\,\alpha_i\cdot\varphi^+\big]\;\; :e^{\beta\alpha_i\cdot\varphi^+}:,$$

with no higher terms, since $\big[J^-,[J^-,\varphi^+]\big]=0$. ■

Remark 5.2 — three caveats, stated honestly

(i) Degree counts fields, not parameters. In C10, T contains $2\alpha^-\partial J^+$: this term carries α^- but lies in $\mathcal{F}^0$. Hence $U_k^- \in \mathcal{F}^{\le 1}$, not "$=\mathcal{F}^1$". Getting this wrong would produce spurious contractions in §7.2.

(ii) The improvement over the relativistic case is a bound, not a category change. A relativistic spin-k current also truncates, at $\le k$ contractions. What is genuinely special here is the drop to $\{0,1,2\}$ and the vanishing of exponent self-contractions.

(iii) The vanishing of self-contractions is the qualitatively new fact. Since $\varphi^+\varphi^+ \sim 0$, the product $\mathscr{S}_i^{(0)}(z)\mathscr{S}_j^{(0)}(w)$ carries no branch factor $(z-w)^{\#}$ from the oscillators: the leading screenings are **oscillator-local**, with no Felder contour prescription needed for that part[Fel89]. Zero-mode cocycles (D4(4)) are a separate question and are not settled by this. This does not extend to the doublet: $\varphi^-\varphi^+ \sim \log$ makes $\mathscr{S}^{(1)}\mathscr{S}^{(0)}$ and $\mathscr{S}^{(1)}\mathscr{S}^{(1)}$ logarithmic, a Takiff/logarithmic extension rather than an ordinary vertex algebra. This is why the screening-screening algebra is deferred (§8.3).

6 Potentials, zero modes, and the module

Convention C7p — the contracted algebra on the plane adopted

§§3 and 7.2 are plane radial-OPE calculations (see the scope decision below), so they need the contracted algebra in plane form. We adopt it explicitly, in exact parallel with C1:

$$J_i^+(z)J_j^+(w) \sim 0, \qquad J_i^-(z)J_j^-(w) \sim 0, \qquad J_i^-(z)J_j^+(w) \sim \frac{\kappa_{ij}/2}{(z-w)^2},$$

together with the plane forms of $T = 2U_2^-$ and $M = 2U_2^+$ of C10. **This is an adoption, not a derivation.** It is the plane counterpart of the cylinder brackets C7 — legitimate because Observation 4.1 shows those brackets are ϵ-exact, so nothing in them is a contraction artefact — but the statement that the plane and cylinder realisations describe the same fields is exactly the bridge that Notebook I does not establish. Marking C7p as adopted is what makes Route B of §7.2 a computation inside one convention system rather than a silent crossing between two.

Definition D3 — Carrollian potentials derived from C7p + W1

Define φ_i^+, φ_i^- by $J_i^\pm = \partial\varphi_i^\pm$. From C7p and W1,

$$\varphi_i^+(z)\varphi_j^+(w) \sim 0, \qquad \varphi_i^-(z)\varphi_j^-(w) \sim 0, \qquad \varphi_i^-(z)\varphi_j^+(w) \sim \tfrac{1}{2}\kappa_{ij}\log(z-w).$$

At $N = 2$ this gives $\varphi^-(z)\varphi^+(w) \sim \frac{1}{4}\log(z-w)$, hence $J^-(z)\varphi^+(w) \sim \frac{1/4}{z-w}$, consistent with $J^+(z)J^-(w) \sim \frac{1/4}{(z-w)^2}$ and with [FH25] (2.11).

Definition D4 — free-field module data (to be completed) derived / open

The word "intertwiner" is meaningless until the following data are fixed. **What is specified here:** the algebraic carrier — index space, oscillators, crossed zero modes, charge lattice and momentum-sector index set. **What is enumerated but not completed here:** the cocycle, the logarithmic residue defining $Q^{(1)}$, the common operator domain, and the proof that the limiting operators are affiliated to the structure. An earlier draft said the module data "are fixed here"; that overstated (4)–(6) below.

What Paper 1 nevertheless requires. Full screening–screening closure may be deferred to Paper 3, but a minimal explicitly defined category and domain — enough to make $Q^{(1)}$ an intertwiner and to state its commutators with $U_k^\pm$ — cannot be. Supplying that minimal category is a Notebook II deliverable, not a Paper 3 one.

(1) Index space. All fields live on the traceless subspace $\mathfrak{h} := \{\lambda \in \mathbb{C}^N : \sum_a \lambda_a = 0\}$, of dimension $N-1$, on which $\kappa = \mathbb{1} - \frac{1}{N}\mathbb{1}\mathbb{1}^\top$ acts as the identity. On all of $\mathbb{C}^N$, κ is a rank-$(N-1)$ projector, $\kappa^2 = \kappa$; the simple roots satisfy $\kappa\alpha_i = \alpha_i$. **The Heisenberg algebra is defined on $\mathfrak{h}$, not on the unconstrained $\mathbb{C}^N$**; Cell 4b verifies the projector properties. Indices are raised and lowered with the induced Euclidean form on $\mathfrak{h}$, so κ^{ij} and κ_{ij} are identified there and only there.

(2) Oscillators. For $n \neq 0$, $J_{i,n}^\pm$ with the brackets of C7, and no others.

(3) Mode expansions, the plane–cylinder map, and crossed zero modes. These must be derived from $J = \partial\varphi$, not guessed, and the two coordinate systems use different mode labels.

Erratum E1 — the mode map, corrected derived

A previous revision of this notebook asserted a mode reversal $a_n = J_{-n}$ between the plane modes $J_{\text{pl}}(z) = \sum_n a_n z^{-n-1}$ and [FH25]'s cylinder modes. **That is false, and it has a one-line falsifier.** The plane OPE $J_{\text{pl}}J_{\text{pl}} \sim \kappa(z-w)^{-2}$ requires $[a_m, a_n] = m\kappa\,\delta_{m+n,0}$; under $a_n = J_{-n}$,

$$[a_m, a_n] = [J_{-m}, J_{-n}] = -m\kappa\,\delta_{m+n,0},$$

the wrong sign. **There is no mode reversal in the unbarred sector.**

What E1 does and does not settle. It settles a negative: no mode reversal. It does **not** fix the bridge, because the overall sign and the placement of [FH25]'s explicit factor of i are separate conventions that must be transported together with φ, α, β, the Miura factors and the barred contour. Two internally consistent candidates illustrate the residual freedom:

$$\text{(i)} \quad z = e^{-i\theta}, \;\; J_{\text{cyl}} = \tfrac{dz}{d\theta}J_{\text{pl}} = -izJ_{\text{pl}} \implies J_n = -a_n;$$

$$\text{(ii)} \quad w = \tau - i\theta, \;\; z = e^w, \;\; J_w = \tfrac{dz}{dw}J_{\text{pl}} = zJ_{\text{pl}}, \;\; J_{\text{FH}} = i\,J_w \implies J_n = +a_n.$$

Candidate (ii) treats [FH25]'s i as belonging to their reality convention $J = i\partial\phi^{\mathrm{FH}}$ rather than to the Jacobian, and reproduces their cylinder dependence $e^{-n\tau+in\theta}$ exactly; it is the leading candidate and is recommended for Notebook II §II.1. **Neither is adopted here.** Both preserve the bracket ($[a_m, a_n] = m\kappa\delta_{m+n,0}$ is invariant under $a_n \to \pm a_n$); only the reversal is excluded. Cell 4c tests all three candidates and rejects the reversal.

What survives regardless of which candidate is chosen. The cylinder expansion below is derived from $\partial_\theta\varphi = i\sum_n J_n e^{in\theta}$ alone, with no plane input, so the zero-mode term $iJ_0\theta$, the monodromy condition, the half-shift and the base-point analysis are unaffected by the residual sign freedom.

Scope decision — two realisations, kept separate

An earlier draft asserted that the whole notebook was formulated on the equal-time cylinder. **That was false**: §3 (parent screening weights, Proposition 3.4) and Route B of §7.2 are plane radial-OPE and residue calculations. The accurate layer statement, which governs the whole notebook, is three-fold:

- **Plane realisation.** The quantum Heisenberg algebra $J_{\mathrm{pl}}J_{\mathrm{pl}} \sim \kappa(z-w)^{-2}$, the quantum Miura realisation, the vertex operators and all residues of §3 and §7.2 are formulated in an independently adopted plane free-field realisation. It is not claimed to be derived from the cylinder data below.
- **Cylinder realisation.** The Carrollian contraction (§4), the averaged ordering (§5) and the zero-mode bookkeeping (§6) are formulated on the equal-time cylinder in [FH25]'s conventions.
- **The bridge between them is NOT asserted here.** Fixing it requires transporting J, φ, α, β, the Miura factors and the barred contour together, which Erratum E1 does not do — E1 establishes only what the bridge cannot be. The complete map is Notebook II §II.1.

What this costs and what it does not. Results proved wholly inside one realisation stand: h_β, $h(\mu\alpha_i)$, $\beta_+\beta_- = -4$ and Proposition 3.4 are plane statements; the contraction identities, the filtration lemma and the averaged-ordering analysis are cylinder statements. **Both routes to F_0 in §7.2 are plane calculations** — Route A contracts a parent plane relation, Route B uses the adopted contracted plane algebra C7p — so their agreement is an internal cross-check inside one adopted realisation. Neither route proves the plane–cylinder bridge, and nothing here licenses an argument that silently crosses between the two realisations.

Cylinder. [FH25] (3.11)–(3.12) fix $J_{\mathrm{cyl}}(\theta) = i\sum_n J_n e^{in\theta}$ with $J = \partial_\theta\varphi$. Integrating term by term,

$$\boxed{\varphi(\theta) = q + i\,J_0\,\theta + \sum_{n\neq 0}\frac{J_n}{n}e^{in\theta}}$$

The zero-mode term is $i\,J_0\theta$, not $J_0\theta$; an earlier draft dropped the i, which changes the monodromy (see layer (ii) of Proposition 6.1). Writing $p_i^\pm := J_{i,0}^\pm$, C7 at $m = 0$ gives no bracket among the p's, so the conjugate pairs are part of the module data, fixed by consistency with $\varphi^-\varphi^+ \sim \frac{1}{2}\kappa\log$ of the adopted plane convention C7p. **This is adopted module data, not a cylinder derivation.** C7 says nothing about the conjugate q-operators: at $m = 0$ its brackets vanish identically. We state the convention rather than appeal to a "standard rule": for $\varphi\varphi \sim A\log(z-w)$ the zero-mode sector must supply the $A\log z$ piece, which requires $[p, q] = A$ (equivalently $[q, p] = -A$). Hence, explicitly, we adopt

$$[\,p_i^-,\ q_j^+\,] = [\,p_i^+,\ q_j^-\,] = \tfrac{1}{2}\kappa_{ij}, \qquad [p^+, q^+] = [p^-, q^-] = [p^+, p^-] = [q^+, q^-] = 0.$$

equivalently $[q_j^+, p_i^-] = [q_j^-, p_i^+] = -\frac{1}{2}\kappa_{ij}$. Within this adopted module convention the factor $\frac{1}{2}$ is fixed by compatibility with D3; it is not inherited through a plane–cylinder bridge, since that bridge is deliberately unresolved. Under $\varphi = i\phi$ (W1) these are i times the real-boson brackets, which is where the reality conditions enter.

(4) Charge lattice and cocycle. By (3), the screening charges shift λ^- by $\frac{\beta_1}{2}\kappa\alpha_i$, so the relevant lattice is

$$\Lambda := \bigoplus_{i=1}^{N-1} \mathbb{Z}\,\tfrac{\beta_1}{2}\kappa\alpha_i \ \subset\ \mathfrak{h},$$

with the factor $\frac{1}{2}$ as in (3). Note Λ is generically not an integral lattice, since β_1 is fixed by F_0 in terms of α^+ and need not be rationally related to the root norms. Regarding the cocycle: because $\varphi^+\varphi^+ \sim 0$, the oscillator part of $\mathscr{S}_i^{(0)}\mathscr{S}_j^{(0)}$ contributes no braiding at all (Remark 5.2(iii)), so **the trivial cocycle $\varepsilon \equiv 1$ may well suffice** — unlike the ordinary lattice-VOA situation, where a nontrivial central extension is forced by the oscillator monodromy. We therefore do not posit a cocycle; we record that the central extension of the Λ-group algebra and the locality requirement it is meant to enforce **must be specified before any statement about mutual locality**, and that the natural first guess here is the trivial one. This is an open item, not a completed definition.

(5) Module, index set and domain. Fix base momenta $(\lambda_0^+, \lambda_0^-) \in \mathfrak{h} \times \mathfrak{h}$ and set

$$\mathcal{H}_{(\lambda_0^+,\lambda_0^-)} := \bigoplus_{\nu\in\Lambda} \mathcal{F}_{\lambda_0^+,\,\lambda_0^-+\nu},$$

an algebraic direct sum (no completion) over the coset $\lambda_0^- + \Lambda$ only — not over all of $\mathfrak{h} \times \mathfrak{h}$ — with $\mathcal{F}_{\lambda^+,\lambda^-}$ the Fock space generated by negative-mode oscillators on a vector of definite (p^+, p^-)-eigenvalue. This index set is closed under the half-shifts of (3)–(4) by construction, and λ^+ is not shifted by the screenings, so it is a fixed label constrained by the integrality condition of Proposition 6.1(ii). Charges act as formal residues on this algebraic module.

Caveat on the residue. $Q_i^{(0)} = \mathrm{Res}_z\, \mathscr{S}_i^{(0)}(z)$ is an ordinary Laurent residue. For the tangent screening this is not available as stated: $\mathscr{S}^{(1)}$ contains φ^-, which pairs logarithmically with φ^+, so $\mathscr{S}^{(1)}$ is not an ordinary Laurent field and $Q^{(1)}$ requires a logarithmic-residue or base-point prescription (cf. layer (iii) of Proposition 6.1). Supplying it is part of the deferred operator layer.

(6) Status, and the split between Notebook II and Paper 3. This is an algebraic induced module. It is not claimed to be a vertex-algebra module in the ordinary sense: by Remark 5.2(iii) the pair $\{\mathscr{S}^{(0)}, \mathscr{S}^{(1)}\}$ is not mutually local (logarithmic $\varphi^-\varphi^+$ pairings), so the ambient category is logarithmic/Takiff. Two different things must not be conflated:

- **Required for Paper 1, and assigned to Notebook II:** a minimal explicitly defined category and common domain — just enough to make $Q^{(1)}$ an intertwiner and to state its commutators with $U_k^{\pm}$. Paper 1 cannot be written without it.
- **Deferred to Paper 3:** the full screening–screening logarithmic algebra, its closure onto the composite charges, and the identification of the complete logarithmic/Takiff category with its braiding and quantum-group structure.

pythonCell 4b — projector properties of κ and the traceless subspace (D4.1)

```
import sympy as sp
for Nv in (2,3,4):
    one = sp.ones(Nv,1)
    kap = sp.eye(Nv) - one*one.T/Nv
    roots = [sp.Matrix([1 if j==i else (-1 if j==i+1 else 0) for j in range(Nv)])
             for i in range(Nv-1)]
    print("N =", Nv,
          "| kappa^2 = kappa:", sp.simplify(kap*kap - kap) == sp.zeros(Nv,Nv),
          "| rank:", kap.rank(), "(= N-1)",
          "| kappa*alpha_i = alpha_i:", all(sp.simplify(kap*r - r) == sp.zeros(Nv,1) for r in roots))
# Cartan matrix in the kappa form
Nv = 3
one = sp.ones(Nv,1); kap = sp.eye(Nv) - one*one.T/Nv
a1 = sp.Matrix([1,-1,0]); a2 = sp.Matrix([0,1,-1])
C = sp.Matrix([[ (x.T*kap*y)[0,0] for y in (a1,a2)] for x in (a1,a2)])
print("\nA_2 Cartan matrix from (alpha_i, kappa alpha_j):"); sp.pprint(C)
```

pythonCell 4c — zero-mode BCH: the half-shift, and the cylinder mode expansion

```
import sympy as sp

# ---- (a) the half-shift.  [p-_i, q+_j] = (1/2) kappa_ij  is CENTRAL, so
#          p exp(X) = exp(X) (p + [p,X]).
b1, ka = sp.symbols('beta_1 (kappa_alpha)_i')
central = sp.Rational(1,2)*ka                 # [p^-_i, q^+_j] alpha_j  ->  (1/2)(kappa alpha)_i
shift   = sp.simplify(b1*central)             # [p^-_i, beta_1 alpha.q^+]
print("momentum shift =", shift, "   (must be beta_1*(kappa alpha)_i / 2, NOT beta_1*(kappa alpha)_i)")
assert sp.simplify(shift - b1*ka/2) == 0
assert sp.simplify(shift - b1*ka) != 0
print("half-shift confirmed; naive full shift excluded")

# ---- (b) cylinder zero mode carries an i.
#      FH (3.11)-(3.12):  J(theta) = i * sum_n J_n exp(i n theta),  and J = d/dtheta phi.
th, J0 = sp.symbols('theta J_0')
n  = sp.Symbol('n', nonzero=True, integer=True)
Jn = sp.Symbol('J_n')
```

```
zero_mode_integrand = sp.I*J0                       # n = 0 term of J(theta)
osc_integrand       = sp.I*Jn*sp.exp(sp.I*n*th)     # n != 0 term
print("\nintegral of n=0 term  :", sp.integrate(zero_mode_integrand, th), "  -> i*J_0*theta")
print("integral of n!=0 term :", sp.simplify(sp.integrate(osc_integrand, th)), "  -> J_n exp(i n theta)/n")
assert sp.simplify(sp.integrate(zero_mode_integrand, th) - sp.I*J0*th) == 0
assert sp.simplify(sp.integrate(osc_integrand, th) - Jn*sp.exp(sp.I*n*th)/n) == 0

# ---- (b2) plane <-> cylinder map: REJECT the mode reversal.  NOTE: this cell
#      certifies ONLY the negative statement a_n != J_{-n}.  The overall sign and the
#      full bridge are NOT fixed in Notebook I; they are Notebook II.1 deliverables.
#      Falsifier: plane OPE requires [a_m,a_n] = m kappa delta_{m+n,0}.
kk = sp.Symbol('kappa', nonzero=True)
def brJ(r, s): return r*kk if sp.simplify(r+s) == 0 else 0     # FH (3.14)
print("\n[a_m,a_-m] under a_n = J_{-n} :", brJ(-m, m), "  (WRONG SIGN -> rejected)")
assert sp.simplify(brJ(-m, m) + m*kk) == 0
print("[a_m,a_-m] under a_n = -J_n   :", brJ(m, -m), "  (correct)")
assert sp.simplify(brJ(m, -m) - m*kk) == 0

#      Coordinate map, with the Jacobian kept:  J_cyl = (dz/dtheta) J_pl
an = sp.Symbol('a_n')
for lab, zz in (("z = exp(+i theta)", sp.exp(sp.I*th)), ("z = exp(-i theta)", sp.exp(-sp.I*th))):
    jac  = sp.diff(zz, th)
    term = sp.simplify(jac*an*zz**(-n-1))
    print("  %s : dz/dtheta = %s z ;  J_cyl term = %s"
          % (lab, sp.simplify(jac/zz), sp.simplify(term/an)))
print("  -> z = exp(-i theta) with THIS Jacobian gives J_n = -a_n;")
print("     assigning FH's explicit i to their reality convention instead gives J_n = +a_n.")
print("     BOTH preserve the bracket; only the reversal a_n = J_{-n} is excluded.")
print("     Notebook I adopts NEITHER sign.  See Erratum E1 and Notebook II.1.")

# ---- (c) monodromy of the leading screening
lamp = sp.Symbol('lambda^+')
mono = sp.simplify(sp.exp(b1*(sp.I*lamp*(th+2*sp.pi)))/sp.exp(b1*sp.I*lamp*th))
print("\nmonodromy factor      :", sp.simplify(mono), " -> single-valued iff beta_1*alpha.lambda^+ in Z")
```

Proposition 6.1 — the three-layer zero-mode dictionary derived (layers i–ii) / formulated here (layer iii)

Status split. Layers (i) and (ii) — the momentum half-shift $\lambda^- \mapsto \lambda^- + \frac{\beta_1}{2}\kappa\alpha_i$ and the monodromy condition $\beta_1\,\alpha_i\cdot\lambda^+ \in \mathbb{Z}$ — are derived within the adopted zero-mode module convention of D4(3), which is chosen data rather than a consequence of C7. Layer (iii), the identification of base-point changes, ordering changes and flag covariance, is a structurally motivated triangular correspondence: the three ambiguities are all strictly lower-triangular on $(\mathscr{S}^{(0)}, \mathscr{S}^{(1)})$, which is what they demonstrably share, but the explicit maps between them are not constructed here. It must not be used as a derived input.

Three logically distinct zero-mode issues are routinely conflated. They must be separated, because they have different resolutions and only the third is genuinely dangerous.

Layer (i) — which momentum is shifted. By D4(3) the conjugate of q^+ is p^-. Hence $\mathscr{S}^{(0)} = e^{\beta\alpha_i\cdot q^+}$ shifts the minus momentum:

$$Q_i^{(0)} \,:\, \mathcal{F}_{\lambda^+,\lambda^-} \longrightarrow \mathcal{F}_{\lambda^+,\,\lambda^- + \frac{\beta_1}{2}\kappa\alpha_i}.$$

The factor $\frac{1}{2}$ is forced and an earlier draft omitted it. Since $[p_i^-,\ \beta_1\alpha_j\cdot q^+] = \frac{\beta_1}{2}(\kappa\alpha)_i$ is central,

$$p_i^-\, e^{\beta_1\alpha\cdot q^+} = e^{\beta_1\alpha\cdot q^+}\Big(p_i^- + \tfrac{\beta_1}{2}(\kappa\alpha)_i\Big),$$

so the shift is $\frac{\beta_1}{2}\kappa\alpha_i$, not $\beta_1\kappa\alpha_i$. Cell 4c verifies this.

The screening charge is therefore an **intertwiner between momentum sectors**, not an endomorphism of a single Fock space; the commutant statement $[Q,U]=0$ is an identity of intertwiners on the coset module $\mathcal{H}_{(\lambda_0^+,\lambda_0^-)} = \bigoplus_{\nu\in\Lambda}\mathcal{F}_{\lambda_0^+,\lambda_0^-+\nu}$ of D4(5) — not an unrestricted sum over all (λ^+,λ^-), which an earlier draft wrote here. This is standard in the Coulomb gas[TK86],[BS93] but must be stated, not assumed.

Layer (ii) — single-valuedness. By D4(3), $\varphi^+(\theta) \supset q^+ + i\,p^+\theta$, so under $\theta\to\theta+2\pi$,

$$\mathscr{S}_i^{(0)}(\theta) \;\longrightarrow\; e^{\,2\pi i\,\beta_1\,\alpha_i\cdot\lambda^+}\;\mathscr{S}_i^{(0)}(\theta), \qquad \text{so single-valuedness} \iff \beta_1\,\alpha_i\cdot\lambda^+ \in \mathbb{Z}.$$

This is a clean integrality condition on the λ^+-support, and it is available only because the i of D4(3) is kept; with the erroneous $q^+ + p^+\theta$ one would obtain a real exponential and no periodicity condition at all. Under $\varphi = i\phi$ (W1) it is the familiar quantisation of a real-boson momentum, which

is why the redefinition cannot be treated as cosmetic. We sidestep the analytic version in Paper 1 by working on the algebraic module of D4, whose architectural precedent is the lattice-extended $\Pi(0) = M(1) \otimes \mathbb{C}[\mathbb{Z}c]$ of Adamović–Radobolja[AR19] — the 2019 continuation, which is where the lattice extension and the logarithmic modules live; [AR16] is the earlier local screening-kernel paper and is the $N = 2$ benchmark of §9, not the module template. That is a rank-one construction with an ordinary (uncrossed) pairing: it is a template for the architecture — lattice extension, intertwining operators, logarithmic modules — and cannot be imported unchanged to rank $N-1$ with the crossed pairing and averaged ordering.

Layer (iii) — base-point dependence of the tangent screening, and its unification with ordering. The tangent prefactor is itself θ-linear — and by D4(3) with the same i as the leading sector, $\varphi^-(\theta) \supset q^- + i\,p^-\theta$ — so $\mathscr{S}^{(1)}$ is quasi-periodic and $Q^{(1)}$ depends on a base point. **This is not a quantum artefact.** The classical bulk charges [DDF26] (5.26c–d) contain undifferentiated $\bar\psi$, and shifting $\bar\psi \to \bar\psi + \text{const}$ shifts S_{κ_2} by a multiple of S_{κ_8}. Simultaneously, reordering $\varphi^- e^{\beta\varphi^+}$ produces only a c-number times $e^{\beta\varphi^+}$, because the crossed commutator is central. Hence three apparently unrelated ambiguities are one and the same:

$$\underbrace{\text{classical base-point shift}}_{\text{[DDF26] } \phi_0} \iff \underbrace{\text{quantum ordering shift}}_{\mathscr{S}^{(1)} \to \mathscr{S}^{(1)} + \lambda \mathscr{S}^{(0)}} \iff \underbrace{\text{flag covariance}}_{\text{§7.3}}.$$

Precision. These three are related by explicit maps, not literally identical, and the paper must exhibit the maps rather than assert the equivalence: the classical base-point shift is a choice of lower limit in $\int_{\phi_0}^{\phi}$; the ordering shift is a c-number determined by a chosen prescription; the flag covariance is the induced ambiguity in a basis of a two-step filtration. Each is a triangular (strictly lower-triangular) transformation of the pair $(\mathscr{S}^{(0)}, \mathscr{S}^{(1)})$, which is what they share.

Lemma 6.2 (Scheme covariance) formulated here

The line $\mathbb{C}\,\mathscr{S}^{(0)} \subset \operatorname{span}\{\mathscr{S}^{(0)}, \mathscr{S}^{(1)}\}$ is independent of the ordering prescription; only the flag, not the individual operator $\mathscr{S}^{(1)}$, is well defined. Consequently the doublet is canonically a two-step filtration, and any statement about $\mathscr{S}^{(1)}$ must be invariant under $\mathscr{S}^{(1)} \to \mathscr{S}^{(1)} + \lambda\mathscr{S}^{(0)}$.

Proof. The basic bracket $[\varphi^-, \varphi^+]$ is a c-number kernel. It does not follow that the commutator of a φ^--mode with a function of φ^+ is central — an earlier draft said so, and that sentence was false. What is true, and is all that is needed, is that the c-number bracket makes the Baker–Campbell–Hausdorff series terminate at first order:

$$\left[\varphi^-,\, e^{\beta\varphi^+}\right] = \beta\,[\varphi^-, \varphi^+]\, e^{\beta\varphi^+},$$

i.e. a scalar kernel multiplying the same exponential — not a central operator, but an element of $\mathbb{C}\,\mathscr{S}^{(0)}$. Hence reordering the prefactor against the exponent changes $\mathscr{S}^{(1)}$ by a scalar multiple of $\mathscr{S}^{(0)}$, and cannot generate $\varphi^+ e^{\beta\varphi^+}$. ■

Corollary. The coefficient B of $\varphi^+ e^{\beta\varphi^+}$ is not an ordering ambiguity: it is determined by the contraction (§7.3). The ordering ambiguity lives entirely in the $C\,\mathscr{S}^{(0)}$ direction.

Caveat — the lemma is conditional, not an unconditional operator theorem. At coincident points the c-number is a formally divergent coincidence limit, so the lemma holds relative to a fixed regularisation (point splitting or mode subtraction), which Notebook I does not select. Centrality guarantees that the mixing is triangular; it does not by itself guarantee that the coefficient is finite or scheme-independent. Fixing the scheme is part of the deferred operator layer.

7 The formal screening jet and the parameter hierarchy

7.1 The jet expansion

We now contract the parent screenings. Potentials contract as the currents do (C6): $\varphi = \epsilon^{-1}\varphi^+ + \epsilon\varphi^-$, $\bar\varphi = \epsilon^{-1}\varphi^+ - \epsilon\varphi^-$. The parent momenta are expanded as follows. Under the hypotheses that (a) the limit is a nonzero local operator in the fixed field φ^+ and (b) the expansion is analytic and odd in ϵ, the leading momentum must scale as $O(\epsilon)$; we adopt

$$\beta = \epsilon\beta_1 + \epsilon^3\beta_3 + O(\epsilon^5), \qquad \bar{\beta} = \epsilon\bar{\beta}_1 + \epsilon^3\bar{\beta}_3 + O(\epsilon^5).$$

Derivation 7.1 — the formal jet derived (formal, commuting symbols)

Status of this computation. What follows is an expansion of commuting exponentials: it computes the associated-graded jet of the screening densities. It does **not** transport normal ordering, the barred sector's radial/mode reversal, zero-mode factors or cocycles. Every statement below is therefore a statement about formal densities; promoting it to a statement about operators is the deferred operator layer and Notebook II.

$$\beta\,\varphi = \beta_1\varphi^+ + \epsilon^2(\beta_1\varphi^- + \beta_3\varphi^+) + O(\epsilon^4), \qquad \bar{\beta}\,\bar{\varphi} = \bar{\beta}_1\varphi^+ + \epsilon^2(-\bar{\beta}_1\varphi^- + \bar{\beta}_3\varphi^+) + O(\epsilon^4).$$

For the two channels to produce a doublet built on the same exponential we need $\bar{\beta}_1 = \beta_1$ — which §7.2 shows is automatic, not an assumption. Then

$$\lim_{\epsilon\to 0} \tfrac{1}{2}\Big(e^{\beta\varphi} + e^{\bar{\beta}\bar{\varphi}}\Big) = \mathscr{S}^{(0)} :=: e^{\beta_1\varphi^+} :,$$

$$\lim_{\epsilon\to 0} \frac{1}{2\epsilon^2}\Big(e^{\beta\varphi} - e^{\bar{\beta}\bar{\varphi}}\Big) = \mathscr{S}^{(1)} :=: \Big(\beta_1\varphi^- + \tfrac{\beta_3-\bar{\beta}_3}{2}\varphi^+\Big)e^{\beta_1\varphi^+} : .$$

So $A = \beta_1$ is **fixed by the contraction** and only $B = \frac{\beta_3-\bar{\beta}_3}{2}$ remains to be determined. Treating A and B as two independent unknowns overparametrises the problem; the honest parameter set is $(\beta_1, \beta_3, \bar{\beta}_3)$.

Proposition 7.2 — classification of the even jet derived

The sum channel also has an $O(\epsilon^2)$ piece, $Q_2 := \oint \frac{\beta_3+\bar{\beta}_3}{2}\,\varphi^+ e^{\beta_1\varphi^+} = \frac{\beta_3+\bar{\beta}_3}{2}\,\partial_{\beta_1} Q_0$, and one might hope for a third screening. An earlier draft of this notebook dismissed it on the grounds that the required subtraction is not itself a parent screening at finite ϵ. **That argument was invalid** — a coefficient in an ϵ-family of commuting operators need not be a parent screening in order to be meaningful. The correct classification is obtained by differentiating the full commutant relation.

Setup. Both parent charges commute with both copies of the currents, so for either channel $[\,Q^{\mathrm{ch}}(\epsilon),\, U_k^\sigma(\epsilon)\,] = 0$ exactly, for all $\epsilon \neq 0$. Expand both factors in ϵ^2:

$$Q^{\mathrm{ch}} = Q_0^{\mathrm{ch}} + \epsilon^2 Q_2^{\mathrm{ch}} + \cdots, \qquad U_k^\sigma(\epsilon) = U_{k,0}^\sigma + \epsilon^2 U_{k,2}^\sigma + \cdots$$

(the currents carry ϵ-corrections because C8 is the $\epsilon \to 0$ limit of (3.16)–(3.17)).

Sum channel. Here $Q_0^{\mathrm{sum}} = Q_0 \neq 0$, so the two lowest orders read

$$O(\epsilon^0):\ \ [Q_0, U_{k,0}^\sigma] = 0, \qquad O(\epsilon^2):\ \ [Q_2, U_{k,0}^\sigma] + [Q_0, U_{k,2}^\sigma] = 0.$$

The second relation is inhomogeneous. It gives $[Q_2, U_{k,0}^\sigma] = -[Q_0, U_{k,2}^\sigma]$ and **nothing more**: both sides may vanish. An earlier draft concluded from it that Q_2 "is not a screening", which does not follow. The defensible statement is conditional:

$$Q_2 \text{ lies in the commutant} \quad\Longleftrightarrow\quad [Q_0, U_{k,2}^\sigma] = 0.$$

What can be settled now. For $\sigma = +$ the condition holds and Q_2 does commute: U_k^+ and all its α^+-derivatives are built solely from J^+, which has no pairing with φ^+, so both terms vanish separately. The question is therefore entirely about $\sigma = -$, where $[Q_0, U_{k,0}^-] = 0$ holds only on the locus F_0 and the compensating term is not known to vanish. **Computing $[Q_0, U_{k,2}^-]$ is a concrete task for Notebook II.** Note what this does and does not count. At the formal coefficient level the difference channel yields two candidate commutant directions, Q_0 and $Q^{(1)}$; their realisation as two nonzero independent operator intertwiners is conditional on Gate 2 of Notebook II — a common finite-ϵ comparison domain, the barred contour and orientation map, a logarithmic-residue or coefficient-limit prescription, the cocycle and zero-mode data, and existence and non-vanishing of the limit.

Saying "at least two independent commutant elements" without that conditional is stronger than anything proved here.

Difference channel. Here the $O(\epsilon^0)$ term cancels identically, so $\frac{1}{2}(Q-\bar{Q}) = \epsilon^2 Q^{(1)} + O(\epsilon^4)$ and the rescaled charge is $Q^{\text{diff}} = Q^{(1)} + O(\epsilon^2)$. The leading order of the commutant relation is therefore homogeneous:

$$O(\epsilon^0): \quad [Q^{(1)}, U^{\sigma}_{k,0}] = 0.$$

Conclusion. The asymmetry between the channels is structural, not accidental: the difference channel starts at order ϵ^2, so rescaling puts its leading commutant equation at homogeneous order, while the sum channel starts at order ϵ^0, so its ϵ^2 coefficient sits at inhomogeneous order. Hence formally, and conditional on (i) existence of the finite-ϵ operator family on a common comparison domain, (ii) the barred contour map selecting this difference channel, and (iii) existence and non-vanishing of the coefficient limit, the contraction delivers **two commutant elements per root** (Q_0 and $Q^{(1)}$), and a possible third, Q_2, whose status is decided by $[Q_0, U^{-}_{k,2}]$. Structurally $Q_2 \propto \partial_{\beta_1} Q_0$ is a momentum/background-charge derivative rather than a new direction — if it does commute, it is a tangent to the screening family, and the honest statement is then that the commutant contains a parameter-derivative direction, not that a third independent screening exists. Either way, the formal multiplicity pattern of §7.4 rests on $Q_0, Q^{(1)}$ alone and is unaffected — pattern, not established commutant multiplicity, which awaits Gate 2.

Caveat. This argument is formal at the level of ϵ-expansions of operators; it presupposes that the expansions exist on a common domain, which is the content of the deferred operator layer below and of Notebook II.

Missing data — contours, orientation, and charges versus densities

Derivation 7.1 adds densities. Adding charges requires data not yet fixed, and the relative sign of the sum/difference channels depends on it:

- the parent charges are $Q = \oint dz\, V(z)$ and $\bar{Q} = \oint d\bar{z}\, \bar{V}(\bar{z})$, the second anti-holomorphic;
- the Carrollian contraction reverses modes in the barred sector, $\bar{J}_{i,-n}$ in C7, which on the cylinder is the map $\theta \mapsto -\theta$ at equal time; the induced orientation reversal of the barred contour contributes a sign to the difference channel;
- [FH25] warn explicitly that reversing radial time $\bar{z} \to 1/z$ acts nontrivially on the u_k because they are not quasi-primary; the same caution applies to $\bar{V}$.

Until the cylinder coordinates, contour orientations and the equal-time mode map are fixed, the labels "sum" and "difference" are placeholders for combinations to be determined. **This does not affect** F_0, which is derived per-copy (Route A) and directly from the Carrollian current (Route B), but it does affect the identification of which combination is $\mathscr{S}^{(0)}$ and which is $\mathscr{S}^{(1)}$, and hence the sign of B. Fixed in Notebook II before any A_2 computation.

7.2 F_0, computed two independent ways

This is the central computation of the notebook. We obtain the leading screening condition twice: once by contracting the relativistic condition, and once by a direct Carrollian OPE. They must agree, and they do.

Route A — contraction of the relativistic condition derived

Insert $\beta = \epsilon\beta_1 + \epsilon^3\beta_3$ and $\alpha = \epsilon^{-1}\alpha^+ + \epsilon\alpha^-$ into $F^{\text{rel}}: \ \beta^2 - 2\alpha\beta - 4 = 0$ (D2, single-field convention):

$$\underbrace{-2\alpha^+\beta_1 - 4}_{O(\epsilon^0)} + \epsilon^2 \underbrace{\left(\beta_1^2 - 2\alpha^+\beta_3 - 2\alpha^-\beta_1\right)}_{O(\epsilon^2)} + O(\epsilon^4) = 0.$$

Hence

$$\boxed{F_0: \quad \alpha^+\,\beta_1 = -2} \qquad \text{and} \qquad F_1: \quad \alpha^+\beta_3 = \tfrac{1}{2}\beta_1^2 - \alpha^-\beta_1.$$

The barred copy has $\bar{\alpha} = \epsilon^{-1}\alpha^+ - \epsilon\alpha^-$, giving at $O(\epsilon^0)$ the identical relation $\alpha^+\bar{\beta}_1 = -2$, so $\bar{\beta}_1 = \beta_1$ automatically — the assumption of Derivation 7.1 is discharged, given $\alpha^+ \neq 0$ and the choice of the same regular branch in both copies — and at $O(\epsilon^2)$, $\alpha^+\bar{\beta}_3 = \frac{1}{2}\beta_1^2 + \alpha^-\beta_1$.

Route B — direct Carrollian OPE derived

Use C10 and D3. For $M = 2J^+J^+ + 2\alpha^+\partial J^+$: no contraction with φ^+ exists, so $M(z)\mathscr{S}^{(0)}(w)$ is **regular** and $[M_n, Q^{(0)}] = 0$ for every β_1. For $T = 4J^+J^- + 2\alpha^+\partial J^- + 2\alpha^-\partial J^+$, using $J^-(z)\varphi^+(w) \sim \frac{1/4}{z-w}$:

- $4J^+J^-$, single contraction via J^-: $4\beta_1 \cdot \frac{1}{4(z-w)} : J^+\mathscr{S}^{(0)} := \frac{1}{z-w}\partial_w\mathscr{S}^{(0)}$. [checked] correct descendant term.
- $2\alpha^+\partial J^-$: $2\alpha^+\beta_1\,\partial_z\frac{1}{4(z-w)} = -\frac{\alpha^+\beta_1/2}{(z-w)^2}$.
- $2\alpha^-\partial J^+$: no contraction. Regular.
- Double contraction: impossible — only one J^- (Lemma 5.1). **No β^2 term.**

$$\Rightarrow \quad h^{(T)}_{\beta_1} = -\frac{\alpha^+\beta_1}{2}, \qquad h^{(T)} = 1 \iff \alpha^+\beta_1 = -2.$$

Proposition 7.3b — F_0 is ordering-independent formulated here

The condition $h^{(T)} = 1$ of Route B is unchanged if U_2^- is taken in normal rather than averaged ordering. Hence F_0 is insensitive to the prescription that distinguishes the symmetric Carrollian algebra from the Galilean one.

Proof. The proposition rests on (i) alone, which is complete. Item (ii) is recorded as a conjectural corroboration only, not as coequal support; its mode-level version belongs to Notebook II. (i) The double pole in $T(z)\mathscr{S}^{(0)}(w)$ arises exclusively from the term $2\alpha^+\partial J^-$, which is linear in the currents and therefore carries no ordering ambiguity whatsoever; the bilinear $4J^+J^-$ can contract only once with $\mathscr{S}^{(0)}$ (Lemma 5.1) and so contributes only to the simple pole. (ii) More generally, the two prescriptions differ on one-J^- products by a c-number, since $[J^-, J^+]$ is central (C7); a c-number differs from zero only in the mode L_0 — this is [FH25]'s own observation that ordering is "only relevant for the zero mode", their (4.23) — and a constant added to L_0 commutes with everything, so every bracket $[L_n, Q]$ is unaffected. Argument (ii) is stated at the level of mode sums and should be written out mode-explicitly before it is relied on; (i) alone suffices for $U_2^\pm$. ■

Scope. This is a statement about $U_2^\pm$. At $N = 3$ the currents $U_3^\pm$ admit two contractions against $\mathscr{S}^{(1)}$, and no analogue of (i) is available; ordering-independence there must be checked, not assumed.

Result 7.3 — the leading screening condition derived; two routes, both inside the adopted plane realisation

$$\boxed{F_0: \quad \alpha^+\,\beta_1^{\text{sf}} = -2 \qquad \Longleftrightarrow \qquad \alpha^+\,\beta_1^{\text{root}} = +1}$$

(the second form via W2, $\beta^{\text{sf}} = -2\mu$, consistent with Proposition 3.3 whose root-basis condition $\mu^2 + \alpha\mu = 1$ contracts to $\alpha^+\mu_1 = 1$). Both forms require $\alpha^+ \neq 0$. Route A and Route B agree exactly.

What the agreement does and does not establish. Both routes are now computations inside the adopted plane realisation (C1 for the parent, C7p for the contracted algebra), so their agreement is a genuine internal cross-check: Route A contracts a parent plane relation, Route B evaluates a plane OPE of the contracted currents, and they are not the same computation. What the agreement does not do is certify that this plane realisation is the image of [FH25]'s cylinder construction — that is the unbridged question of §6, and it is why C7p is marked adopted rather than derived. An earlier draft called these "two independent exact routes" without that qualification. The condition is **linear in** β_1 — there is no β^2 term, because exponent self-contractions vanish — but it is a **reciprocal relation** between α^+ and β_1, not a statement that the quantum shift has disappeared.

Interpretation 7.4 formulated here — the background charge is the divergent branch

F_0 is the contracted remnant of the relativistic reciprocity $\beta_+\beta_- = -4$. Concretely: if β_+ is the regular branch (the full single-field root, not the Toda coupling $b = \beta_+/2$), with $\beta_+ = \epsilon\beta_1 + O(\epsilon^3) \to 0$, then the dual branch $\beta_- = -4/\beta_+ \sim -4/(\epsilon\beta_1)$ diverges. Since $\alpha = \frac{1}{2}(\beta_+ + \beta_-)$, the finite Carrollian background charge is

$$\alpha^+ = \lim_{\epsilon\to 0} \epsilon\,\alpha = \lim_{\epsilon\to 0} \frac{\epsilon}{2}\left(\epsilon\beta_1 - \frac{4}{\epsilon\beta_1}\right) = -\frac{2}{\beta_1},$$

which is exactly F_0. **Physical statement:** in the Carrollian limit the two Coulomb-gas screenings separate completely — one survives as a local screening operator, and the other is entirely converted into the background charge. The "$b + 1/b$" quantum structure is not lost; it is re-encoded as the reciprocal relation $\alpha^+\beta_1 = -2$. Anyone expecting the naive statement "the Carrollian background charge is unshifted" is misreading the limit; anyone expecting the shift to survive as an additive $1/b$ is also misreading it.

Falsifiable content and its exact scope. If a second local screening per root existed at leading order, this interpretation would fail. Within the class of limits considered here — regular local limits on the algebraic Fock module of D4 — the divergence of β_- forbids it, since no scalar rescaling $r(\epsilon)$ can make $e^{\beta_-\varphi}$ finite when the exponent itself blows up. **This does not exclude distributional or nonregular limits**, in which an exponential of a divergent argument may retain meaning; that is precisely the regime of [FPT26], and it is flagged as a possible route to a second sector in Paper 3 rather than excluded.

pythonCell 5 — F₀ and F₁ by contraction; agreement with the direct OPE; the divergent-branch identity

```
import sympy as sp
e   = sp.Symbol('epsilon', positive=True)
apS, amS = sp.symbols('alpha^+ alpha^-')
B1, B3, B3b = sp.symbols('beta_1 beta_3 beta_3bar')

def expand_condition(sexp, aexp, order=4):
    """F_rel = beta^2 - 2*alpha*beta - 4, expanded in epsilon (beta = single-field momentum)."""
    F = sp.expand(sexp**2 - 2*aexp*sexp - 4)
    return sp.series(F, e, 0, order).removeO().expand()

# --- unbarred copy
beta_un  = e*B1 + e**3*B3
a_un  = apS/e + e*amS
F_un  = expand_condition(beta_un, a_un)
# --- barred copy
beta_bar = e*B1 + e**3*B3b
a_bar = apS/e - e*amS
F_bar = expand_condition(beta_bar, a_bar)

F0     = sp.expand(F_un.coeff(e, 0))
F1_un  = sp.expand(F_un.coeff(e, 2))
F1_bar = sp.expand(F_bar.coeff(e, 2))
print("O(eps^0) [unbarred] :", F0, "= 0   ->  F0")
print("O(eps^0) [barred]   :", sp.expand(F_bar.coeff(e,0)), "= 0   (identical -> beta_1bar = beta_1)")
print("O(eps^2) [unbarred] :", F1_un, "= 0")
print("O(eps^2) [barred]   :", F1_bar, "= 0")
print()

# solve
sol_b1  = sp.solve(sp.Eq(F0,0), B1)[0]
sol_b3  = sp.solve(sp.Eq(F1_un ,0), B3 )[0]
sol_b3b = sp.solve(sp.Eq(F1_bar,0), B3b)[0]
print("F0  ->  beta_1  =", sol_b1, "      i.e.  alpha^+ * beta_1 = -2")
print("F1  ->  beta_3  =", sp.simplify(sol_b3))
print("F1b ->  b3bar   =", sp.simplify(sol_b3b))
print()

# --- Route B: direct Carrollian OPE gave h = -(alpha^+ beta_1)/2 ; set h = 1
routeB = sp.Eq(-apS*B1/2, 1)
print("Route B condition :", routeB, " -> beta_1 =", sp.solve(routeB, B1)[0])
print("Routes A and B agree:", sp.simplify(sp.solve(routeB,B1)[0] - sol_b1) == 0)
print()

# --- Interpretation 7.4 : alpha^+ from the divergent dual branch
bp = e*B1                      # regular branch
bm = -4/bp                     # dual branch, from beta_+ beta_- = -4
ap_from_branches = sp.simplify(sp.limit(e*(bp + bm)/2, e, 0))
print("alpha^+ from branches :", ap_from_branches,
      "   == -2/beta_1 ?", sp.simplify(ap_from_branches + 2/B1) == 0)
```

7.3 F_1, and the determination of B

Derivation 7.5 — the tangent coefficient derived

Subtracting the two $O(\epsilon^2)$ relations of Route A,

$$\alpha^+(\beta_3 - \bar\beta_3) = -2\alpha^-\beta_1 \quad\Longrightarrow\quad B = \frac{\beta_3 - \bar\beta_3}{2} = -\frac{\alpha^-\beta_1}{\alpha^+} \overset{F_0}{=} \frac{\alpha^-\beta_1^2}{2}.$$

This is a **formal jet-coefficient relation**: it becomes a statement about an operator coefficient only after the operator layer is discharged. Hence, up to overall normalisation and modulo the flag ambiguity of Lemma 6.2,

$$\boxed{\mathscr{S}^{(1)} \;\propto :\Big(\varphi^- - \frac{\alpha^-}{\alpha^+}\,\varphi^+\Big)e^{\beta_1\varphi^+}:}$$

In particular, **if** $\alpha^- = 0$ **the tangent screening is purely** $\varphi^- e^{\beta_1\varphi^+}$, with no φ^+ prefactor.

Resolution R1 — the bulk locus is $\alpha^- = 0$ derived (was open item O1)

An earlier draft left this open. It is not open, and three independent lines of evidence agree.

(i) Central charges. Three-dimensional asymptotically flat Einstein gravity has $c_L = 0,\ c_M = 3/G \neq 0$ for $\mathfrak{bms}_3$. Comparing with C9,

$$c_L^{\mathrm{cl}} = -4(N-1)N(N+1)\alpha^+\alpha^- = 0, \qquad c_M^{\mathrm{cl}} = -2(N-1)N(N+1)(\alpha^+)^2 \neq 0$$

forces $\alpha^- = 0$ with $\alpha^+ \neq 0$. This is the whole of it: the flat-space locus is the vanishing of one of the two contracted background charges, and $\alpha^+ \neq 0$ is exactly the condition F_0 already needs.

(ii) Structure of the bulk charges. At $\alpha^- = 0$ Derivation 7.5 gives $B = -\alpha^-\beta_1/\alpha^+ = 0$, so the tangent screening is purely $\varphi^- e^{\beta_1\varphi^+}$, with no φ^+ prefactor — which is precisely the form of [DDF26] (5.26c–d), whose prefactors are built only from barred fields. The absence of an unbarred prefactor in the bulk is therefore explained, not merely observed.

(iii) Primary source, now verified verbatim. [DDF26] states it explicitly. Their general Miura formula (6.51) is followed by footnote 9, which reads: "In [32], this corresponds to $\alpha^- = 0$." — reference [32] being [FH25]. The body text gives the reason in their own terms: their expression coincides with [FH25]'s "in the case where one of the central charges vanishes, $c_1 = 0$". That is precisely line (i) above: their c_1 is our c_L. The earlier draft recorded this as unconfirmed; it is confirmed, and lines (i)–(ii) are independent confirmations of it rather than substitutes.

Consequences, now mandatory downstream. The Carrollian screening problem relevant to flat higher-spin holography sits at $\alpha^- = 0$; there $c_L = 0$, $B = 0$, F_1 collapses to $\alpha^+\beta_3 = \alpha^+\bar\beta_3 = \frac{1}{2}\beta_1^2$, and the formal density doublet is $\{\, e^{\beta_1\varphi^+},\ \varphi^- e^{\beta_1\varphi^+} \,\}$ — densities, not yet operators: existence, contour definition and the coefficient limit remain Notebook II gates. The general-α^- formulas are retained above because the algebra exists off the bulk locus and the paper should state the general result, but every comparison with [DDF26] must be made at $\alpha^- = 0$.

Dictionary D5 — bulk-side source data and the open normalisation matching problem verbatim transcription + open matching

The same passage that settles $\alpha^- = 0$ also fixes most of the dictionary that Theorem 5 needs, and it was not used in the previous revision. In [DDF26], the symbol θ in (6.49) is the nilpotent dual-number generator, not the angular coordinate used elsewhere in this notebook. To prevent a genuine notation collision, we rename that source symbol ϑ here and impose explicitly

$$\vartheta^2 = 0, \qquad \vartheta \neq \theta_{\mathrm{cyl}}.$$

With this notational renaming only, [DDF26] identify the diagonal entries as

$$D_{jj} = \frac{1}{\sqrt{\pi\kappa\alpha_s}}\Big(i\,\vec\epsilon_j\cdot\vec\psi\,' + i\,\vartheta\,\vec\epsilon_j\cdot\vec{\bar\psi}\,'\Big), \qquad \vec\epsilon_i\cdot\vec\epsilon_j = \delta_{ij} - \frac{1}{s}, \quad \sum_j \vec\epsilon_j = 0,$$

and write the invariant operator as $(\alpha\partial + i\vec\epsilon_1\cdot\vec\psi\,')\cdots$ with $\alpha = \sqrt{\pi\kappa\alpha_s}$, $\alpha_s = 12/(s(s^2-1))$, and (flat space) $\kappa = 1/(8\pi G)$. Three consequences:

- **Their weight pairing is our** κ**.** $\vec\epsilon_i\cdot\vec\epsilon_j = \delta_{ij} - \frac{1}{s}$ is literally C1's κ_{ij}. No translation needed.
- **The background charge is fixed only up to a matching constant.** [DDF26] say their operator coincides with [FH25]'s "up to normalization factors". We therefore do **not** identify the two directly. Introduce matching constants and solve for them:

$$\alpha^+_{\rm FH} = c_\alpha\,\alpha_{\rm DDF}, \qquad J^+_j = c_J(-i\,\vec{\epsilon}_j\cdot\vec{\psi}\,'), \qquad J^-_j = c_-(-i\,\vec{\epsilon}_j\cdot\vec{\psi}\,'),$$

with (c_α, c_J, c_-) determined by matching four things: the crossed brackets C7 against [DDF26] (5.23)/(6.23); the coefficients of the Miura operator; c_M against the bulk Newton constant; and the chosen real form. **Until that system is solved, no numerical value of α^+ is claimed.** An earlier draft asserted $\alpha^+ = 1/(4\sqrt{G})$ and $\mu_1 = 4\sqrt{G}$; those are withdrawn.

Real-form diagnostic — conditional, not an obstruction theorem conditional inference

Status first. An earlier draft presented this as a derived obstruction. It is not: the argument presupposes that the two central generators have already been matched with positive real relative normalisation, which is exactly what D5 leaves open. In general $c_M^{\rm FH} = \lambda\, c_2^{\rm DDF}$ with λ part of the unsolved matching problem, and the sign and phase of λ are what the conclusion turns on. The correct statement is conditional:

If the [FH25] and [DDF26] central generators are identified with positive real relative normalisation ($\lambda > 0$), then α^+ is imaginary in the [FH25] real form and c_α is imaginary.

Under that hypothesis, from C9 at the rank relevant to Paper 1 and to the [DDF26] higher-spin comparison, $N = 3$, $c_M = -2(N-1)N(N+1)(\alpha^+)^2 = -48(\alpha^+)^2$ (consistent with the spin-three value $\alpha_s = \frac{1}{2}$); and three-dimensional asymptotically flat gravity has $c_M > 0$ (in [DDF26]'s normalisation (4.23), $c_2 = 3/G$), so

$$(\alpha^+)^2 = -\frac{c_M^{\rm FH}}{48} = -\frac{\lambda\, c_2^{\rm DDF}}{48} < 0 \qquad \Longrightarrow \qquad \alpha^+ \text{ is imaginary in the [FH25] convention,}$$

the denominator being the $N = 3$ coefficient of C9, rather than the distinct $N = 2$ coefficient. With [DDF26]'s displayed normalisation $c_2^{\rm DDF} = 3/G$ this reads $(\alpha^+)^2 = -\lambda/(16G)$. **No numerical value is claimed:** only under the additional choice $\lambda = 1$ would this give $\alpha^+ = \pm i/(4\sqrt{G})$, and λ is precisely what the open matching problem must supply. What survives without λ is the sign conclusion, for any $\lambda > 0$ real.

whereas $\alpha_{\rm DDF} = \sqrt{\pi\kappa\alpha_s}$ is manifestly real in the displayed classical parametrisation, giving an imaginary c_α — a real-form rotation rather than a mere rescaling. This is a useful diagnostic, consistent with (and suggestive of the same i as) the $\varphi = i\phi$ bridge of W1 seen from the gravity side, but it is not yet forced. Note also that $F_0 :\ \alpha^+\beta_1 = -2$ then makes β_1 imaginary too, so that $\beta_1\varphi^+ = \beta_1(i\phi^+)$ is real — internally consistent. **The precise numerical factor is normalisation-dependent** (it turns on how c_M versus c_2 are normalised between the two papers) and must come out of the matching system, not from an assumed identification.

Root-pattern identification — not a completed root-to-exponent map. What is established is which exponent combinations occur and how they are organised by the root system; the exact correspondence $\beta_1\alpha_i\cdot\varphi^+ \leftrightarrow -\psi_{(L)} \mp 2\psi_{(W)}$ still requires the normalisation matrix and real-form constants that Notebook IV is tasked to determine. With $\alpha_i = \vec{\epsilon}_i - \vec{\epsilon}_{i+1}$, the exponents of the pure-gauge parameters (5.16) are $-\psi_{(L)} \mp 2\psi_{(W)}$ for the two simple roots and $-2\psi_{(L)}$ for the highest root $\alpha_1 + \alpha_2$ — and indeed [DDF26] (5.16a) contains precisely $\epsilon_{(W)}^{-2} = \kappa_1 e^{-2\psi_{(L)}}$. This also explains the composite charges: the highest-root constants are κ_1, κ_7, which are exactly the labels of the two composite charges (5.27a–b). The Takiff assignment of Derivation 8.1 is thereby corroborated from the bulk side.

Status summary for D5, so that it cannot conflict with O1′ or Observation 7.6: established — the pairing $\vec{\epsilon}_i\cdot\vec{\epsilon}_j = \kappa_{ij}$, and the root pattern (two simple-root exponents plus a highest-root exponent $-2\psi_{(L)}$, matching the constants κ_1, κ_7 of the composite charges). Open — the normalisation constants (c_α, c_J, c_-), the real-form map, the explicit basis matrix reconciling $8\pi G$ with $6\pi G$, and therefore every numerical value and the exact root-to-exponent correspondence. These are O1′ and Theorem 5, assigned to Notebook IV.

What genuinely remains open here — O1′

Not whether $\alpha^- = 0$, but the exact dictionary at that locus: the map from $(\psi_{(L)}, \psi_{(W)})$ to the root basis $\alpha_i\cdot\varphi^+$; the reconciliation of the two bulk normalisations $8\pi G$ (spin two) and $6\pi G$ (spin three)

in [DDF26] (5.23) with the single κ_{ij} of C1; the overall constants κ_n; and the identification of α^+ with G, α_s. This is Theorem 5.

pythonCell 6 — FORMAL associated-graded jet (commuting symbols); coefficients A, B

```
import sympy as sp
e = sp.Symbol('epsilon', positive=True)
pp, pm = sp.symbols('varphi^+ varphi^-')          # COMMUTING symbols: this is a FORMAL
# associated-graded jet. It cannot see BCH terms, cocycles, zero modes, or averaged ordering.
B1, B3, B3b = sp.symbols('beta_1 beta_3 beta_3bar')
apS, amS = sp.symbols('alpha^+ alpha^-')

phi     = pp/e + e*pm
phibar  = pp/e - e*pm
beta_un    = e*B1 + e**3*B3
beta_bar   = e*B1 + e**3*B3b

Eun  = sp.series(sp.exp(beta_un*phi ),  e, 0, 4).removeO().expand()
Ebar = sp.series(sp.exp(beta_bar*phibar), e, 0, 4).removeO().expand()

S0   = sp.simplify(sp.limit((Eun + Ebar)/2, e, 0))
diff = sp.simplify(sp.expand((Eun - Ebar)/(2*e**2)))
S1   = sp.simplify(sp.limit(diff, e, 0))
summ = sp.simplify(sp.limit(sp.expand((Eun + Ebar)/2 - sp.exp(B1*pp))/e**2, e, 0))

print("S^(0) =", sp.simplify(S0))
print("S^(1) =", sp.factor(sp.simplify(S1/sp.exp(B1*pp))), " * exp(beta_1 phi^+)")
print("sum-channel O(eps^2) piece =", sp.factor(sp.simplify(summ/sp.exp(B1*pp))), " * exp(...)")
print()

# now impose F1 to eliminate beta_3, beta_3bar
b3_sol  = sp.solve(sp.Eq(B1**2 - 2*apS*B3  - 2*amS*B1, 0), B3 )[0]
b3b_sol = sp.solve(sp.Eq(B1**2 - 2*apS*B3b + 2*amS*B1, 0), B3b)[0]
A_coef  = B1
B_coef  = sp.simplify((b3_sol - b3b_sol)/2)
print("A =", A_coef, "   (fixed by the contraction)")
print("B =", sp.simplify(B_coef), "  = -alpha^- beta_1 / alpha^+ ?",
      sp.simplify(B_coef + amS*B1/apS) == 0)
print("with F0 (alpha^+ = -2/beta_1):  B =", sp.simplify(B_coef.subs(apS, -2/B1)))
print("normalised tangent operator:  (phi^-  +  (B/A) phi^+) exp(beta_1 phi^+),  B/A =",
      sp.simplify(B_coef/A_coef))
```

7.4 Branch counting: 8 → 4

Observation 7.6 formulated here — the multiplicity and organisation match the bulk (evidence)

Count the relativistic A_2 screenings: 2 simple roots $\times$ 2 Toda branches $\times$ 2 chiral copies $= 8$. The contraction keeps one branch per root (Interpretation 7.4) and folds the two chiral copies into the jet doublet, giving

$$2 \text{ roots } \times \ 1 \text{ regular branch } \times \ 2 \text{ jet components } \ = \ 4,$$

which is exactly the number of elementary Carrollian screening charges found in the bulk, [DDF26] (5.26a–d): two pure exponentials and two polynomial-times-exponential partners. Their ∓ 2 pattern in $e^{-\psi_{(L)} \mp 2\psi_{(W)}}$ is plausibly the two simple roots of A_2 written in the bulk weight basis, but that identification requires the explicit weight-basis map and the reconciliation of the $8\pi G$ and $6\pi G$ normalisations of [DDF26] (5.23); it is a target (Theorem 5), not an established dictionary.

Status discipline. This predicts the number of jet components; it does not prove the existence of two quantum screening operators, which is Gate 2 of Notebook II. It is a multiplicity and organisation match, and belongs in the introduction as evidence. It is not a derivation: it does not fix the root dictionary, normalisations, relative signs, which branch survives in the bulk frame, or possible cocycle sectors. Those are Theorem 5, and **Notebook IV**.

8 The bulk targets

8.1 The classical Carrollian Coulomb gas

Convention C13 — bulk Poisson structure verbatim [DDF26] (5.23)

$$\{\psi'_{(L)}(\phi), \bar{\psi}_{(L)}(\phi')\} = \{\bar{\psi}'_{(L)}(\phi), \psi_{(L)}(\phi')\} = 8\pi G\, \delta(\phi - \phi'),$$

$$\{\psi'_{(W)}(\phi), \bar{\psi}_{(W)}(\phi')\} = \{\bar{\psi}'_{(W)}(\phi), \psi_{(W)}(\phi')\} = 6\pi G\, \delta(\phi - \phi').$$

Convention C14 — the four elementary bulk screening charges verbatim [DDF26] (5.26a–d)

$$S_{\kappa_8} = \frac{\kappa_8}{8\pi G}\int d\phi\; e^{-\psi_{(L)}-2\psi_{(W)}}, \qquad S_{\kappa_9} = \frac{\kappa_9}{8\pi G}\int d\phi\; e^{-\psi_{(L)}+2\psi_{(W)}},$$

$$S_{\kappa_2} = -\frac{\kappa_2}{8\pi G}\int d\phi\; (\bar\psi_{(L)} + 2\bar\psi_{(W)})e^{-\psi_{(L)}-2\psi_{(W)}},$$

$$S_{\kappa_3} = -\frac{\kappa_3}{8\pi G}\int d\phi\; (\bar\psi_{(L)} - 2\bar\psi_{(W)})e^{-\psi_{(L)}+2\psi_{(W)}}.$$

Convention C15 — the two composite charges rewrite of [DDF26] (5.27a–b)

Writing $e_\mp := e^{-\psi_{(L)}\mp 2\psi_{(W)}}$ and $\bar\chi_\mp := \bar\psi_{(L)} \pm 2\bar\psi_{(W)}$,

$$S_{\kappa_7} = -\frac{\kappa_7}{2\pi G}\int d\phi\Big[e_-\int_{\phi_0}^{\phi} d\phi' e_+ - e_+\int_{\phi_0}^{\phi} d\phi' e_-\Big],$$

$$S_{\kappa_1} = -\frac{\kappa_1}{2\pi G}\int d\phi\Big[e_-\bar\chi_-\int_{\phi_0}^{\phi} d\phi' e_+ - e_+\bar\chi_+\int_{\phi_0}^{\phi} d\phi' e_-\Big].$$

These are obtained from Poisson brackets of the elementary four, and carry an explicit base point ϕ_0.

8.2 The Takiff grading, and which bracket makes which composite

Structural inference 8.1 — sector-by-sector matching inference from the field content of C13–C15

Status. What follows is an argument from field content and the crossed Poisson pairing, not an evaluated bracket. Promoting it to "derived" requires writing at least one representative Poisson bracket in full, with its signs, the primitive of the delta function, the lower limit ϕ_0, and the constants κ_n. That is a Notebook IV task; the corroboration from the bulk root structure in D5 is independent supporting evidence, not a substitute.

C13 pairs ψ only with $\bar\psi$; there is no $\psi\psi$ or $\bar\psi\bar\psi$ pairing. Three consequences follow immediately, and they line up exactly with the quantum structure of §5.3:

- **Leading × leading vanishes.** $S_{\kappa_8}, S_{\kappa_9}$ depend only on ψ, so $\{S_{\kappa_8}, S_{\kappa_9}\} = 0$. The quantum counterpart requires care and an earlier draft overstated it: what follows unconditionally from $\varphi^+\varphi^+ \sim 0$ (Remark 5.2(iii)) is that **the oscillator parts of the leading screenings commute** — their OPE is regular, with no anomalous branch power. The charge-level statement $[Q_i^{(0)}, Q_j^{(0)}] = 0$ additionally requires the zero-mode group-algebra cocycle of D4(4) to be trivial, or to be a compatible commuting central extension; D4(4) explicitly leaves that open. The correct reading is therefore: the oscillator parts commute; with the trivial cocycle the integrated leading charges commute. This matters most for the two A_2 simple-root intertwiners, where compatibility of the two cocycle sectors is precisely what must be checked.
- **Tangent × leading** $\to S_{\kappa_7}$**.** The single $\bar\psi$ of the tangent charge is consumed by the bracket, leaving a purely exponential expression; the nested $\int_{\phi_0}^{\phi}$ arises because C13 pairs $\bar\psi$ with ψ', so recovering $\{\bar\psi, \psi\}$ requires integrating a delta — which is precisely where the base point enters.
- **Tangent × tangent** $\to S_{\kappa_1}$**.** One $\bar\psi$ is consumed, one survives, giving the polynomial-times-exponential nested structure.

This is the correct Takiff pattern for the pair, and it corrects a natural but wrong guess that both composites come from tangent × leading.

8.3 Why the composite layer is deferred

Deferral O2 — with precedent

Quantising the elementary four is one problem; proving that their commutators close onto quantum versions of C15 is a second, harder one, involving logarithmic kernels, ordered contours, periodic Green functions, contact terms and base-point dependence. The precedent is decisive:

in the structurally simpler Heisenberg–Virasoro/$W(2,2)$ setting, the local screening of Adamović–Radobolja[AR16] and the nonlocal bosonic formula for it[AR19] — with explicit logarithmic kernels, and a conjectural Kazhdan–Lusztig-dual quantum group — required two separate papers. We therefore treat this as Paper 3, not as a gate on Paper 1.

9 The problem, precisely stated

What Notebooks II, III and IV must establish

Notebook II gate (A_1 operator layer) — all seven of: the barred equal-time map and contour orientation with no unresolved sign; a common algebraic domain or explicit embeddings I_ϵ; $Q^{(0)}$ finite and nonzero with an explicit momentum-sector map; $Q^{(1)}$ finite and nonzero with a stated base point and flag covariance; the A_1 commutators reproducing F_0 exactly; F_1, B, multiplicative normalisation and triangular C-mixing separately identified; and the even jet classified. Passing only the first three supports a leading-screening paper, not a doublet paper.

Notebook III gate (A_2/W_3) — items (a)–(c) below, on an explicitly named module, with all singular OPE terms displayed, every nonzero singular part written as a total derivative, no unexplained contact term, no hidden ordering change, the exact parameter locus stated, and the two simple-root computations cross-checked against one another.

Notebook IV gate (bulk match) — items (d)–(e) below: the matching constants c_α, c_+, c_-, the $8\pi G$ vs $6\pi G$ reconciliation by an explicit basis matrix, the $G \to 0$ scaling with what is held fixed, and the four-charge match.

(a) Lemma 5.1 at A_2 with $U_3^\pm$, in the primary basis $\mathcal{U}_3^+ = U_3^+ - \frac{\alpha^+}{2}\partial U_2^+$, $\mathcal{U}_3^- = U_3^- - \frac{\alpha^-}{2}\partial U_2^+ - \frac{\alpha^+}{2}\partial U_2^-$.
(b) $[Q_i^{(a)}, U_{k,n}^\sigma] = 0$ for $k = 2,3$, $a = 0,1$, $\sigma = \pm$, on the module of Proposition 6.1, in averaged ordering.
(c) The full hierarchy F_0, F_1 at A_2, including whether the classical relations of §7.2 receive quantum corrections from the averaged-ordering reordering terms.
(d) The dictionary to C13–C14 at the locus $\alpha^- = 0$, completing D5: the root-to-(L,W) map, the $8\pi G$ vs $6\pi G$ normalisation non-uniformity, and the constants κ_n (item O1′).
(e) The $N = 2$ cross-checks against [AR16] and against the Carrollian Liouville couplings of [HVZ26].

Exactly what §7 has and has not established

An earlier draft called both routes to F_0 "essentially classical". **That was wrong**, and the correction matters because it changes which question is open.

What is established. F_0 is a quantum relation on both routes: Route A contracts a condition obtained from the normal-ordered parent stress tensor, and Route B is a radial-ordered OPE with the quantum current U_2^-. By Proposition 7.3b it is moreover ordering-independent. Therefore, once the vertex convention (W1) is fixed, $F_0 : \ \alpha^+\beta_1 = -2$ is exact as a **parameter relation**, subject only to $\alpha^+ \neq 0$.

What is not established, and is the actual content of Theorem 4. Three things, none of which is a statement about F_0 as an equation:

- **Existence.** That the parent vertex operators possess a limit as averaged-ordered operators on a common module — including finite multiplicative normal-ordering renormalisations, cocycles, and the zero-mode domain. Until this is done, $\mathscr{S}^{(0)}, \mathscr{S}^{(1)}$ are formal densities, not operators (operator layer; D4(4)–(5)).
- **F_1.** Unlike F_0, the subleading relation involves the tangent operator, whose ordering is not immaterial. $B = -\alpha^-\beta_1/\alpha^+$ is so far a formal jet coefficient, and what is open is whether the tangent operator exists as an averaged-ordered limit and with what multiplicative normalisation. **It cannot receive a triangular scheme shift**: by Lemma 6.2 the ordering ambiguity moves $\mathscr{S}^{(1)} \to \mathscr{S}^{(1)} + \lambda\mathscr{S}^{(0)}$ and therefore changes only the C-direction, never the coefficient of $\varphi^+ e^{\beta_1\varphi^+}$. The earlier draft asserted the opposite and contradicted its own lemma.
- **$N = 3$.** Whether the $U_3^\pm$ singular parts are total derivatives, where the one-J^- filtration permits two contractions and averaging is genuinely operative.

javascriptCell 7 — scratchpad (JS): numerical sanity check of the branch limit

```
// numerical illustration of Interpretation 7.4:
// alpha = (beta_+ + beta_-)/2 with beta_+ beta_- = -4 ; alpha^+ = lim eps*alpha
const b1 = 1.7;                    // choose beta_1
let out = [];
```

```
for (const eps of [1e-1, 1e-2, 1e-3, 1e-4]) {
  const bp = eps*b1;               // regular branch
  const bm = -4/bp;                // dual branch
  const alpha = (bp + bm)/2;
  out.push(`eps=${eps.toExponential(0)}  beta_+=${bp.toExponential(3)}  ` +
           `beta_-=${bm.toExponential(3)}  eps*alpha=${(eps*alpha).toFixed(8)}`);
}
out.push(`target -2/beta_1 = ${(-2/b1).toFixed(8)}`);
return out.join("\n");
```

10 Audit ledger

Every equation used above, its provenance, and how it was checked. Anything not on this list is not licensed for use in the paper.

Item	Source / origin	Status	Check
$[J_{i,m}, J_{j,n}] = m\kappa_{ij}\delta_{m,-n}$	[FH25] (3.14)	verbatim	vs (2.10) at $N = 2$ — Check 2.1
$\sum_i J_i = 0$	[FH25] (3.4)	verbatim	—
Miura operator; u_2; c^{cl}	[FH25] (3.1), (3.5), (3.6)	verbatim	Cell 1, $N = 2, 3, 4$
$T =: JJ : +\alpha\partial J$, $c = 1 - 6\alpha^2$	[FH25] (2.8)–(2.10)	verbatim	Check 2.2
$\varphi_i\varphi_j \sim +\kappa_{ij}\log$	W1, from the adopted plane OPE of C1	derived from an adopted convention	descendant term in Der. 3.1; not obtained from the classical cylinder bracket
$h_\beta = \frac{\beta^2}{4} - \frac{\alpha\beta}{2}$	Derivation 3.1	derived	Cell 2
$\beta_+\beta_- = -4$; $c = 1 - 6(b - 1/b)^2$	Derivation 3.2	derived	Cell 2 (reproduces standard result)
A_{N-1} simple-root conventions $\alpha_i = e_i - e_{i+1}$	C5; [FL88], [BS93] for context	imported	root dictionary W2 checked; the screening theorem itself is now Prop. 3.4, not imported
Contraction C6–C7; Miura C8; T, M, c_L, c_M C9; $N = 2$ C10	[FH25] (3.7)–(3.10), (3.13), (3.15), (1.1)–(1.2), (3.28)–(3.34)	verbatim	Cell 4
ϵ-exactness of the oscillator brackets (no representation claim)	Observation 4.1	derived	Cell 3 (δ-support only)
Symmetry of crossed pairing; $m/4$ at $N = 2$	Check 4.2	derived	Cell 3
Averaged ordering; one-J^- filtration	[FH25] (2.14), (4.29)–(4.34), §4.4	verbatim	—
Truncation bounds $0, 1, 1, 2$; BCH termination	Lemma 5.1	formulated here	proof from C7 + C12 only
Zero-mode half-shift $\lambda^- \to \lambda^- + \frac{\beta_1}{2}\kappa\alpha_i$; monodromy $\beta_1\alpha_i \cdot \lambda^+ \in \mathbb{Z}$	Prop. 6.1, layers (i)–(ii)	derived within the adopted module convention D4(3)	Cell 4c
Base-point / ordering / flag-covariance correspondence	Prop. 6.1, layer (iii)	formulated here	triangular structure exhibited; explicit maps not constructed
Scheme covariance (flag)	Lemma 6.2	formulated here	centrality of $[\varphi^-, \varphi^+]$
$A = \beta_1$ fixed by the contraction (formal jet)	Derivation 7.1	derived (formal)	Cell 6

Item	Source / origin	Status	Check
Even jet: commutes iff $[Q_0, U^-_{k,2}] = 0$; automatic for $\sigma = +$	Proposition 7.2	derived (formal, conditional)	open for $\sigma = -$; Notebook II. Multiplicity claims are conditional on operator existence.
F_0 : $\alpha^+ \beta_1^{\mathrm{sf}} = -2$	Result 7.3, two routes	derived	Cell 5 (A vs B agree)
F_1; $B = -\alpha^- \beta_1 / \alpha^+$	Derivation 7.5	derived (formal jet only)	Cell 6; an operator coefficient only after Notebook II
Background charge = divergent branch	Interpretation 7.4	formulated here	Cell 5, Cell 7
Branch counting $8 \to 4$	Observation 7.6	formulated here	vs [DDF26] (5.26) multiplicity
Bulk Poisson structure (5.23) and the four elementary charges (5.26a–d)	[DDF26]	verbatim	transcription only
The two composite charges, in the abbreviations $e_\mp, \bar{\chi}_\mp$	C15 from [DDF26] (5.27a–b)	rewrite	notation introduced here
Bulk locus $\alpha^- = 0$	[DDF26] fn. 9 after (6.51); + C9 with $c_L = 0$	verbatim + derived	Resolution R1; quoted and cross-derived
[DDF26] (6.49)–(6.51) as printed: $\alpha_{\mathrm{DDF}} = \sqrt{\pi\kappa\alpha_s}$, $\vec{\epsilon}_i \cdot \vec{\epsilon}_j = \delta_{ij} - \frac{1}{s}$	[DDF26] (6.49)–(6.51)	verbatim	transcription only
Identification of those with $\alpha^+, \kappa_{ij}, J^+_j$	Dictionary D5	matching ansatz	source says ”up to normalization factors”
Root pattern: two simple-root exponents + highest root $e^{-2\psi_{(L)}}$	D5, vs [DDF26] (5.16a), (5.27)	derived	pattern only; the exact root↔exponent map needs the normalisation matrix (Notebook IV)
Zero-mode half-shift $\lambda^- \to \lambda^- + \frac{\beta_1}{2}\kappa\alpha_i$; cylinder $i\,p\,\theta$; monodromy $\beta_1 \alpha_i \cdot \lambda^+ \in \mathbb{Z}$	D4(3)–(4), Prop. 6.1(i)–(ii)	derived	Cell 4c; corrects a factor-2 error
Commutant: inclusion only, not equality	programme box, Theorem 3	bounded	equality needs a kernel/character theorem
Missing: logarithmic residue for $Q^{(1)}$	D4(5) caveat	open	$\mathscr{S}^{(1)}$ is not a Laurent field
Plane–cylinder map: **no mode reversal**, $a_n \neq J_{-n}$	Erratum E1	derived (negative statement only)	Cell 4c falsifies the reversal by bracket sign. **The overall sign is NOT fixed here**; both $J_n = \pm a_n$ survive. Bridge = Notebook II.1.
FH25 (3.3) as printed omits m; (3.14) contains it. We cite (3.14)	Source note S1	verbatim	confirmed by direct inspection of the canonical v3 PDF (digest in the manifest)
Real-form diagnostic: α^+ imaginary, c_α imaginary; $c_M = -48(\alpha^+)^2$ at $N = 3$	D5 diagnostic box	conditional inference	holds only if $c_M^{\mathrm{FH}} = \lambda c_2^{\mathrm{DDF}}$ with $\lambda > 0$ real; λ is open

Item	Source / origin	Status	Check
Open: matching constants c_α, c_J, c_-; all numerical values of α^+	D5, O1′	open	Theorem 5; no numbers claimed
Composite-sector assignment (tangent×leading, tangent×tangent)	Structural inference 8.1	inference	needs one explicit Poisson bracket
Field redefinition $\varphi = i\phi$, $J = i\partial\phi$	W1, from [FH25] fn. 3 + (3.11)–(3.12)	rewrite	reality conditions stated; momenta transported
Plane quantum OPE $J_i J_j \sim \kappa_{ij}(z-w)^{-2}$	C1	adopted	[FH25]'s quantum normalisation; not a rewrite of the classical cylinder bracket
Contracted algebra in plane form; plane T, M	C7p	adopted	plane counterpart of C7; legitimacy rests on Observation 4.1's ϵ-exactness
General-root weight $h = \mu^2 + \alpha\mu$	Proposition 3.3	derived	Cell 2b, all simple roots, $N \leq 5$
Total-derivative witnesses at $N = 3$, both simple roots	witness box, §3.3	derived	Cell 2d; residuals vanish identically on the locus
$[Q_i^{(0)}, Q_j^{(0)}] = 0$ at charge level	Structural inference 8.1	conditional	oscillator parts commute unconditionally; charge level needs the trivial/compatible cocycle (D4(4))
Same μ screens all u_k: $\mathrm{Res}[u_k V] = (\mu^2 + \alpha\mu - 1)\mathcal{R}_{k,i}$	Proposition 3.4 (+ C3q ordering)	derived	Cell 2c, trace-aware $\mathcal{D}$, genuine divisibility test, $N = 2, 3, 4$, all simple roots
Reserved notation: b Toda coupling vs β momentum	§3.2	rewrite	$\beta_+ = 2b$; $b^2 - \alpha b - 1 = 0$
F_0 ordering-independent	Proposition 7.3b	formulated here	two proofs; scope limited to $U_2^\pm$
Traceless index space; coset index set	D4(1),(5)	derived	Cell 4b (projector)
Crossed zero-mode algebra $[p_i^-, q_j^+] = \frac{1}{2}\kappa_{ij}$	D4(3)	adopted module data	compatible with C7p; not determined by C7, whose $m = 0$ brackets vanish
Cocycle ε and the logarithmic residue for $Q^{(1)}$	D4(4), D4(5) caveat	open	module data are not complete; Notebook II
Bulk locus $\alpha^- = 0$; $B = 0$; $c_L = 0$	Resolution R1	derived	from C9 + flat-space $c_L = 0$; [DDF26] (5.26c–d)
Hypothesis: $\alpha^+ \neq 0$	required by F_0	hypothesis	stated wherever F_0 is used
Hypothesis: both parent momenta lie in the regular $O(\epsilon)$ branch/scaling class	§7.1 scaling ansatz	hypothesis	this is where the assumption sits
$\bar{\beta}_1 = \beta_1$ inside that class	§7.2, Route A at $O(\epsilon^0)$	derived	forced given $\alpha^+ \neq 0$; not a separate assumption

Item	Source / origin	Status	Check
Missing data: contour orientation, cylinder-mode map for $\bar{Q}$	"Missing data" box, §7.1	open	fixes the sign of B; Notebook II
Missing data: zero-mode cocycle ε	D4(4)	open	not determined by the oscillator sector
Existence of the limiting operators (vs. formal densities)	Derivation 7.1 status note; §9	open	Notebook II — Theorem 4
Quantum corrections to F_1 from averaging	§9	open	Notebook II — Theorem 4
Averaging corrections at $N = 3$ (nonlinear $U_3^{\pm}$)	§9	open	Notebook III — Theorem 3

Reproducibility and offline verification rewrite

This interactive page fetches fonts, MathJax, Pyodide and SymPy from CDNs and therefore is not, by itself, an archival rendering. The package now also ships `Notebook_I_Setup_Static.pdf`, a static rendering that requires no network or browser runtime. **What is mechanically certified is the source-and-verifier package and the byte identity of that static companion, not visual equivalence between the interactive and static renderings.** The versioned, checkable objects are the HTML source, static PDF, offline verifier, mechanical freeze check, shared divisibility predicate, mutation regression, pinned environment and manifest. ("Frozen" is reserved for $G_{0,\mathrm{sci}}$; see the gate statement below.) Seven companion components travel with the notebook:

- `Notebook_I_Setup_Static.pdf` — a self-contained static rendering for archival reading. It contains no executable notebook runtime; the HTML source and offline Python verifier remain the computational artifacts of record.
- `verify_notebook_I.py` — a standalone verifier, no browser and no network, exiting nonzero on any failure. **Its scope is exactly:** the embedded SymPy identity checks (classical Miura polynomials, root-weight algebra, oscillator brackets on the δ-support, the $N = 2$ contraction, F_0/F_1, the zero-mode half-shift, the projector identities) and the formal commuting-symbol jet checks. **It does verify**, within its adopted symbolic model, the $N = 3$ total-derivative witnesses (§10b) and the cylinder monodromy formula (§8). **It does not verify** source transcription, independent validation of the Wick/OPE implementation, averaged ordering, the filtration proof, cocycles, contour construction, operator domains, the even-jet commutant question, or the bulk dictionary. Those are prose or open.
- A pinned environment for the authoritative check: Python $\geq$ 3.10 with **SymPy 1.14.0**, enforced — the verifier now exits nonzero on a version mismatch (override with `--allow-any-sympy` for exploratory use only). **The in-page runtime is not pinned:** the browser loads whatever SymPy is bundled with Pyodide 0.26.4, which need not be 1.14.0. The embedded cells are therefore illustrative and convenient; the offline verifier is the artefact of record, and any discrepancy between them is resolved in favour of the verifier.
- `SOURCE_MANIFEST.sha256` — SHA-256 fingerprints. **Actively hash-checked lines:** all eight distributed artifacts — this HTML, the static PDF, `verify_notebook_I.py`, `freeze_check.py`, `divisibility.py`, `test_freeze_gate_mutations.py`, `requirements.txt` and `ROADMAP.md`. **Not package-checked:** the three primary-source PDFs. They are ordinary arXiv-generated PDFs with text layers; their SHA-256 digests have been independently recomputed by two separate parties and agree exactly with the values recorded in the manifest. They remain provenance comments rather than active entries for one reason only: **third-party source files are not redistributed as part of this package**. If a freeze bundle does include them, promote the three comment lines to active hash lines.
- `divisibility.py` — the single divisibility predicate used by the **offline gate components** (the browser cell reimplements the test for convenience), with positive, negative and contract controls.
- `test_freeze_gate_mutations.py` — a negative control for the gate itself. It breaks the shared predicate three ways: an always-true return, replacement of the cancelled quotient by the unevaluated quotient R/F, and removal of the polynomiality rejection. It then stages the **whole package** and applies nine single-component bypasses: five in the verifier (including valid early-green and disabled-accumulator cases), two in the freeze checker, and two in the mutation runner. The

manifest is regenerated after each mutation, so the test measures behavioural detection rather than a hash mismatch. Return code, a distinct completion sentinel and the reported PASS count must all agree. Every subprocess is bounded, and independent package mutations are evaluated concurrently in isolated temporary directories.

- `freeze_check.py` — a behavioural freeze gate. Its predecessor searched for reassuring strings and was defeated by a mutation that kept the strings while disabling the tests; it now runs the controls, self-tests its own result accumulator, executes the verifier in ordinary and negative-control modes, validates their sentinels and check counts, validates the mutation runner's distinct sentinel and exact case count, and performs document-control scans (which are legitimately textual). A caller-side recursion guard prevents staged child gates from launching the mutation suite again.

The gate. In a clean extracted directory, all four of

```
sha256sum -c SOURCE_MANIFEST.sha256
python3 verify_notebook_I.py
python3 test_freeze_gate_mutations.py
python3 freeze_check.py
```

must return zero. This is $G_{0,\mathrm{mech}}$, the mechanical gate. It is not the scientific freeze:

$$G_{0,\mathrm{mech}} = \text{manifest} + \text{verifier} + \text{mutation regression} + \text{freeze checker},$$
$$G_{0,\mathrm{sci}} = G_{0,\mathrm{mech}} + \text{human scientific sign-off}.$$

The direct mutation-regression command and the freeze checker are deliberately cross-witnessed: the former mutates the latter, while the latter rejects a silent or sentinel-forging mutation runner. **Only $G_{0,\mathrm{sci}}$ authorises the word "frozen".** The four commands returning zero establish $G_{0,\mathrm{mech}}$ and nothing more. **What the gate certifies is bounded:** document control, regression against previously-found defects, the behavioural divisibility controls, and the successful execution of the verifier and mutation regression. The verifier certifies only the symbolic identities it lists — not source transcription, operator existence, averaged ordering, cocycles, contour signs, or the bulk normalisation map. **A green gate is not a scientific freeze certificate for anything the ledger marks open. Cell numbering.** In-page cells run 0, 1, 2, 2b, 2c, 2d, 3, 4, 4b, 4c, 5, 6 in document order — twelve Python cells — plus one JavaScript scratchpad; the lettered cells were inserted in revision and the numbering is kept stable rather than renumbered, so that references in the audits remain valid.

11 References

[FH25] S. Fredenhagen and L. Hörl, Free-field construction of Carrollian W_N-algebras, arXiv:2509.14441v3 [hep-th]; JHEP **02** (2026) 218. Primary source for the current and Miura conventions of §§2, 4, 5. Author and journal data confirmed via [DDF26] ref. [32].

[DDF26] E. D'Arcy, A. Delfante and S. Fredenhagen, Holographic realization of higher-spin Carrollian free fields, arXiv:2604.27068v1 [hep-th]. Primary source for §8 and D5; eqs. (5.16a), (5.23), (5.26a–d), (5.27a–b), (6.23), (6.27), (6.49)–(6.51) and footnote 9. Authors verified: Ethan D'Arcy, Arnaud Delfante, Stefan Fredenhagen.

[FPT26] S. Fredenhagen, S. Prohazka and R. Tiefenbacher, Carrollian quantum states and flat space holography, arXiv:2604.22745v2 [hep-th]. Nonregular states; relevant to Prop. 6.1 layer (ii) and to Paper 3.

[FL88] V. A. Fateev and S. L. Lukyanov, The models of two-dimensional conformal quantum field theory with Z_n symmetry, Int. J. Mod. Phys. A **3** (1988) 507. Parent screening currents; the statement imported in C5.

[DF84] Vl. S. Dotsenko and V. A. Fateev, Conformal algebra and multipoint correlation functions in 2D statistical models, Nucl. Phys. B **240** (1984) 312. Coulomb-gas background charge and screening.

[BS93] P. Bouwknegt and K. Schoutens, W-symmetry in conformal field theory, Phys. Rept. **223** (1993) 183, arXiv:hep-th/9210010. Standard reference for Miura, screenings, Fock resolutions.

[Fel89] G. Felder, BRST approach to minimal models, Nucl. Phys. B **317** (1989) 215. Contour/BRST structure that Remark 5.2(iii) shows is not needed for the leading sector.

[TK86] A. Tsuchiya and Y. Kanie, Fock space representations of the Virasoro algebra — intertwining operators, Publ. RIMS **22** (1986) 259. Screenings as intertwiners; Prop. 6.1 layer (i).

[AR16] D. Adamović and G. Radobolja, On free field realizations of $W(2,2)$-modules, SIGMA **12** (2016) 113, arXiv:1605.08608 [math.QA]. $N = 2$ benchmark: the local screening operator whose kernel is the $W(2,2)$ vertex algebra. The $\Pi(0)$ module template is [AR19], not this paper.

[AR19] D. Adamović and G. Radobolja, Self-dual and logarithmic representations of the twisted Heisenberg–Virasoro algebra at level zero, Commun. Contemp. Math. **21** (2019) 1850008, arXiv:1703.00531 [math.QA]. Nonlocal bosonic screening formula with logarithmic kernels (their eq. (7.26)); **source of the lattice-extended module $\Pi(0) = M(1) \otimes \mathbb{C}[\mathbb{Z}c]$ used as the architectural template in D4**; precedent for the deferral O2; BMS_3 contact at $c_L = 26$.

[AR15] D. Adamović and G. Radobolja, Free field realization of the twisted Heisenberg–Virasoro algebra at level zero and its applications, J. Pure Appl. Algebra **219** (2015) 4322, arXiv:1405.1707 [math.QA].

[HVZ26] D. Hidalgo, S. Vandoren and H. Zeng, BMS_3 invariant field theories, arXiv:2607.06826 [hep-th]. Electric/magnetic/canonical BMS_3 scalars and Carrollian limits of Liouville theory; action-level $N = 2$ anchor.

[Rad21] G. Radobolja, Galilean W_3 algebra, J. Math. Phys. **62** (2021) 081701. Galilean (flipped) counterpart — the algebra our novelty claim must be distinguished from.

Reference hygiene — status by item

Equation-level verified against the primary PDF. [FH25]: (1.1)–(1.4), (2.8)–(2.14), (3.1)–(3.18), (3.23), (3.26)–(3.34), (3.36)–(3.41), (4.1), (4.20)–(4.34), §4.4. [DDF26]: (5.23), (5.26a–d), (5.27a–b), (6.19)–(6.27), (6.33)–(6.37).

Corrected and verified in this revision. The initials of [FH25] and [FPT26] were previously wrong ("M. Hörl", "M. Tiefenbacher"). **"L. Hörl" is now verified**: [DDF26] cites the paper as its reference [32], "S. Fredenhagen and L. Hörl, Free-field construction of Carrollian W_N-algebras, JHEP 02 (2026) 218 [2509.14441]", which also confirms the journal data. [DDF26]'s own authors are likewise confirmed as Ethan D'Arcy, Arnaud Delfante and Stefan Fredenhagen. "R. Tiefenbacher" is confirmed by independent review against the title page of arXiv:2604.22745v2.

No longer load-bearing. [FL88] was previously the sole external dynamical input of Paper 1. That dependence is **eliminated** by Proposition 3.4, which derives the parent screening identity directly in the conventions C1/C3 for $N \leq 4$, covering the $N = 3$ case of Paper 1. [FL88] and [BS93] are now cited for the origin of the general-N result and for context only, and their bibliographic details need routine verification rather than an equation-level pin. [DF84], [BS93], [Fel89], [TK86], [AR15], [Rad21] are contextual and require routine bibliographic verification.

Now verified. [DDF26] footnote 9, attached to eq. (6.51): "In [32], this corresponds to $\alpha^- = 0$." Together with (6.49)–(6.51) this supplies the bulk-side fields, Miura factors and screening data from which D5 is to be solved — not the completed dictionary, whose constants c_α, c_+, c_- and $8\pi G/6\pi G$ basis reconciliation remain open (Notebook IV). Verified against arXiv:2604.27068v1 on 1 August 2026, and independently confirmed by direct inspection of the canonical PDF whose SHA-256 is recorded in the manifest.

Source manifest. Record the SHA-256 and version dates of the exact [FH25] and [DDF26] PDFs used, so that equation numbers can be re-checked against the same versions. [FH25] is cited at v3.

Notebook I of the Carrollian screening programme (Volume I of IV). Volume II: the operator layer and the A_1 commutation theorem in averaged ordering on the module of D4. Volume III: the A_2/W_3 commutation theorem with $U_3^\pm$ in the primary basis. Volume IV: holographic matching at $\alpha^- = 0$ and the $N = 2$ cross-checks. The authoritative programme map, gates and risk register are in `ROADMAP.md`, which travels with this package under the manifest. All scientific content, choices of direction, and responsibility for correctness rest with the author; this notebook is a computational and organisational instrument.

JavaScript is disabled. Mathematics will not render and the executable cells will not run. The mathematical content is present in the page source; a standalone offline verifier (`verify_notebook_I.py`) reproduces the embedded symbolic identities without a browser; see the reproducibility section for its exact scope.